

FIRST TERM – JSS 2 INTERMEDIATE SCIENCE LESSON NOTES



WEEK 1: REVIEW OF JSS 1 SCIENCE CONCEPTS

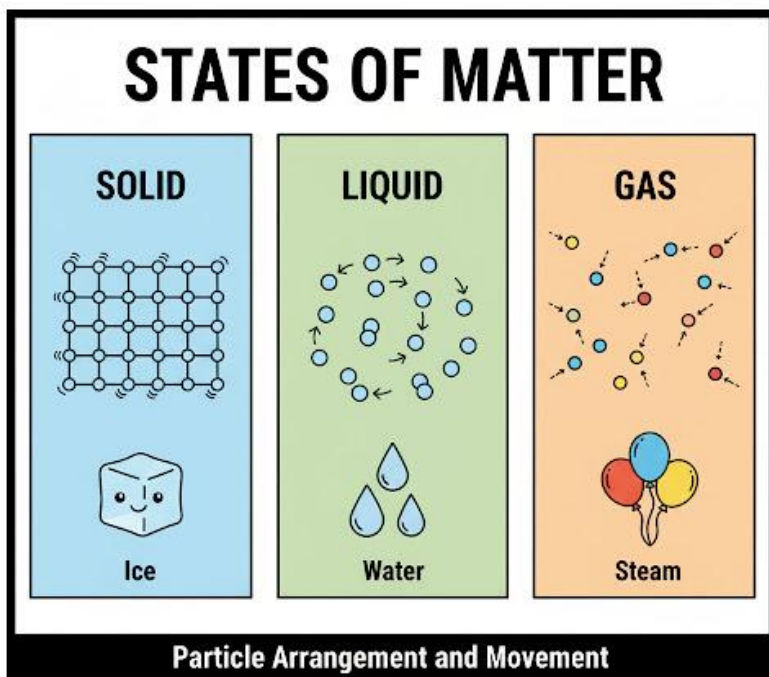
Topic: Review of JSS 1 Science Concepts

Breakdown (Subtopics):

- Revision of matter, energy, and environment
- Laboratory rules and safety recap
- Importance of observation and measurement

Performance Objectives: By the end of this lesson, students should be able to:

1. Define matter and list its three states with examples.
2. Define energy, list five forms, and state the law of energy conservation.
3. Define the environment and its basic components.
4. List at least ten essential laboratory rules.
5. Explain the importance of observation and measurement in science.



Lesson Content:

1. Revision of Matter

- **Definition:** Matter is anything that has **mass** (weight) and occupies **space** (volume).
- **States of Matter:**
 - **Solid:** Has a definite shape and definite volume. The particles are tightly packed and vibrate in fixed positions (e.g., stone, ice, wood, book).
 - **Liquid:** Has no definite shape (takes the shape of its container) but has a definite volume. The particles are close but can slide past one another (e.g., water, kerosene, milk).
 - **Gas:** Has no definite shape and no definite volume (fills the entire container). The particles are far apart and move randomly and quickly (e.g., air, oxygen, steam).
- **Properties of Matter:**
 - **Physical Properties:** Can be observed without changing the substance (e.g., colour, size, density, melting point, boiling point).
 - **Chemical Properties:** Describe how a substance reacts with other substances (e.g., rusting of iron, burning of wood).

2. Revision of Energy

- **Definition:** Energy is the **ability to do work** or cause change.
- **Forms of Energy:**
 - **Potential Energy:** Stored energy (e.g., a rock on a hill, a stretched rubber band).
 - **Kinetic Energy:** Energy of motion (e.g., a rolling rock, a moving car).
 - **Heat Energy:** Energy from the vibration of particles (e.g., fire, stove).
 - **Light Energy:** Energy we can see (e.g., sunlight, bulb).
 - **Sound Energy:** Energy from vibrations we can hear.
 - **Chemical Energy:** Energy stored in bonds (e.g., food, batteries, fuel).
 - **Electrical Energy:** Energy from the flow of electrons (e.g., electricity).
- **Law of Conservation of Energy:** "Energy can neither be created nor destroyed, but it can be converted (changed) from one form to another."

- *Example:* A television converts Electrical Energy into Light Energy and Sound Energy.

3. Revision of Environment

- **Definition:** The environment is the total surroundings of a living organism, including all the factors that can influence it.
- **Components:**
 - **Biotic (Living):** Plants, animals, fungi, bacteria.
 - **Abiotic (Non-living):** Air, water, soil, sunlight, temperature, wind.

4. Laboratory Rules and Safety Recap The science laboratory is a place for experiments, but it can be dangerous. Safety is the first priority.

1. NEVER enter the laboratory without a teacher's permission.
2. NEVER eat, drink, or chew gum in the laboratory.
3. Read all instructions *before* starting an experiment.
4. Wear protective gear (safety goggles, lab coats) when instructed.
5. Report ALL accidents, spills, or breakages to the teacher *immediately*.
6. NEVER taste or smell chemicals directly.
7. Always point test tubes *away* from yourself and others when heating.
8. Keep your workspace clean and tidy.
9. Do not play, run, or shout in the laboratory.
10. Wash your hands thoroughly before and after every experiment.

5. Importance of Observation and Measurement Science is built on evidence, not guesses.

- **Observation:** The process of using your five senses (sight, hearing, smell, taste, touch) to gather information about the world.
 - *Qualitative Observation:* Describes qualities (e.g., "The liquid turned blue," "The gas smells bad").
- **Measurement:** The process of using scientific instruments to get exact, *quantitative* data (numbers).

- *Importance:* Measurement is more accurate and reliable than just observation.
- *Instruments:*
 - Metre Rule (for Length)
 - Weighing Balance/Scale (for Mass)
 - Measuring Cylinder/Beaker (for Volume)
 - Stopwatch (for Time)
 - Thermometer (for Temperature)

Evaluation:

1. What is matter? List its three states.
2. State the Law of Conservation of Energy.
3. What is the difference between a biotic and an abiotic component of the environment?
4. List five important laboratory safety rules.
5. What is the difference between a qualitative observation and a quantitative measurement?

Assignment:

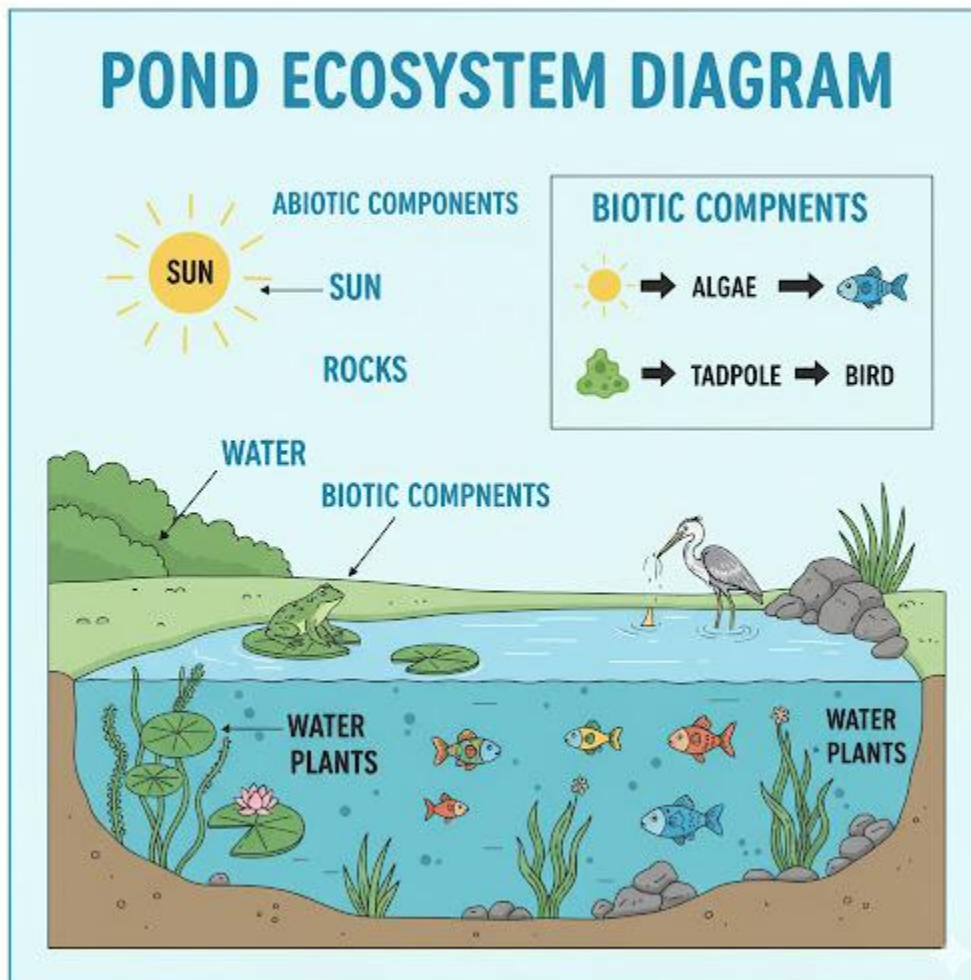
1. Define (a) Physical Property and (b) Chemical Property. Give one example of each.
2. List five different forms of energy and give an example of a device that converts energy (e.g., a fan).
3. Why should you never eat or drink in the laboratory?
4. List five instruments used for measurement in science and state what each one measures.
5. What is the difference between "observation" and "inference"? (e.g., Observation: The ground is wet. Inference: It must have rained).

WEEK 2: COMPONENTS OF THE ENVIRONMENT

Topic: Components of the Environment

Breakdown (Subtopics):

- Biotic and abiotic components
- Interdependence between living and non-living things
- Food chains and food webs
- Energy flow in ecosystems



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Ecosystem."

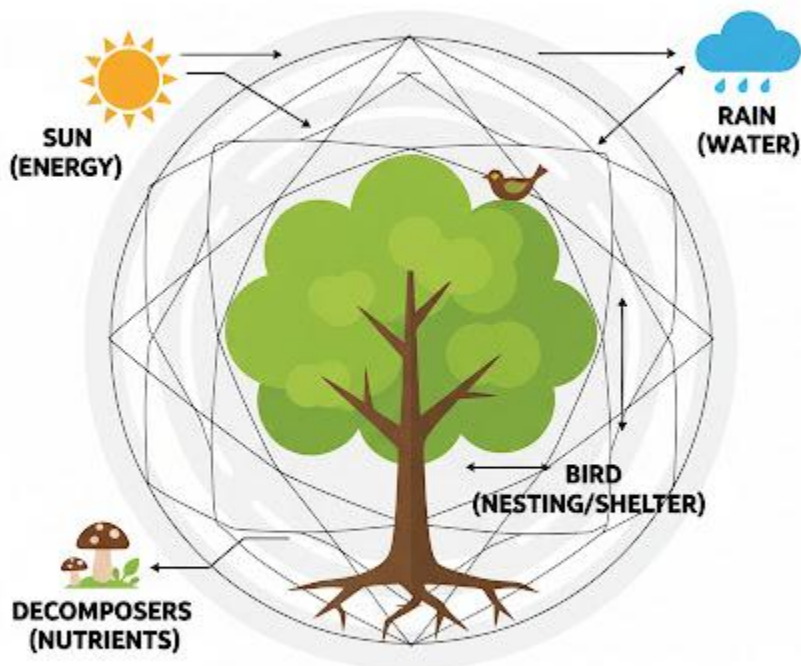
2. List and give examples of biotic and abiotic components.
3. Explain "interdependence" with at least two examples.
4. Draw and label a simple food chain.
5. Define and differentiate between producers, consumers, and decomposers.

Lesson Content:

1. What is an Ecosystem?

An **Ecosystem** is a community of living organisms (biotic) interacting with their non-living (abiotic) environment. An ecosystem can be as large as a forest or as small as a pond.

INTERDEPENDENCE



All Living Things Are Connected In A Cycle of Life.

2. Components of an Ecosystem

- **A. Biotic (Living) Components:** All the living parts of an ecosystem.
 - **Producers (Autotrophs):** Organisms that make their own food, usually using sunlight. All **green plants** are producers (e.g., grass, trees, algae).
 - **Consumers (Heterotrophs):** Organisms that get energy by eating other organisms.
 - *Primary Consumers (Herbivores):* Eat producers (e.g., grasshopper, cow, goat).
 - *Secondary Consumers (Carnivores/Omnivores):* Eat primary consumers (e.g., frog, lion).
 - *Tertiary Consumers:* Eat secondary consumers (e.g., snake, hawk).
 - *Omnivores:* Eat both plants and animals (e.g., humans, chickens, bears).
 - **Decomposers:** Organisms that break down dead plants and animals and their wastes (e.g., faeces) into simple nutrients, returning them to the soil. This is a vital role.
 - *Examples:* **Bacteria** and **Fungi** (like mushrooms).
- **B. Abiotic (Non-living) Components:** All the non-living parts of an ecosystem.
 - **Sunlight:** The main source of energy for *all* ecosystems.
 - **Water (Rainfall):** Essential for all life (drinking, photosynthesis).
 - **Air:** Provides oxygen (for respiration) and carbon dioxide (for photosynthesis).
 - **Soil (Land):** Provides nutrients, water, and an anchor for plants; a habitat for animals.
 - **Temperature:** Affects what kind of plants and animals can live in an area.

3. Interdependence Interdependence means that all components of an ecosystem (biotic and abiotic) *depend on each other* for survival. No organism can live alone.

- **Example 1:** A plant (biotic) needs sunlight, water, and soil (abiotic) to grow.
- **Example 2:** An animal (biotic) needs air (abiotic) to breathe and a plant (biotic) for food.
- **Example 3:** A decomposer (biotic) breaks down a dead animal (biotic), returning nutrients to the soil (abiotic), which are then used by a plant (biotic).

4. Food Chains and Food Webs

- **Food Chain:** Shows a *single pathway* of how energy (food) flows in an ecosystem.
 - *Example:* Sun → Grass → Goat → Lion (Energy) (Producer) (Primary Consumer) (Secondary Consumer)
- **Food Web:** More realistic. It is a network of *many interconnected food chains* showing that most animals eat more than one thing.
 -
 - *Example:* A hawk might eat a frog, a snake, *and* a rat. A snake might eat a frog *or* a rat.

5. Energy Flow in Ecosystems

- Energy flow is **one-way**. It starts from the **Sun**.
- Producers (plants) capture about 1% of the sun's energy.
- When a herbivore eats a plant, it only gets about **10%** of the plant's energy. (The other 90% is lost as heat or used for the plant's life processes).
- When a carnivore eats the herbivore, it only gets **10%** of the herbivore's energy.
- This is why there are very few "top predators" (like lions) but a lot of producers (like grass).

Evaluation:

1. What is an ecosystem?
2. List three biotic components and three abiotic components.
3. What is the difference between a primary consumer and a secondary consumer?
4. What is the important role of decomposers like bacteria and fungi?
5. Draw a simple food chain with three levels (producer, primary consumer, secondary consumer).

Assignment:

1. Define Producer, Consumer, and Decomposer.

2. What is the difference between a food chain and a food web?
3. Explain the concept of "interdependence" using a plant, an animal, and water.
4. Draw a simple food web (at least 5 organisms).
5. What is the original source of all energy in almost every ecosystem? Why is energy "lost" at each level?

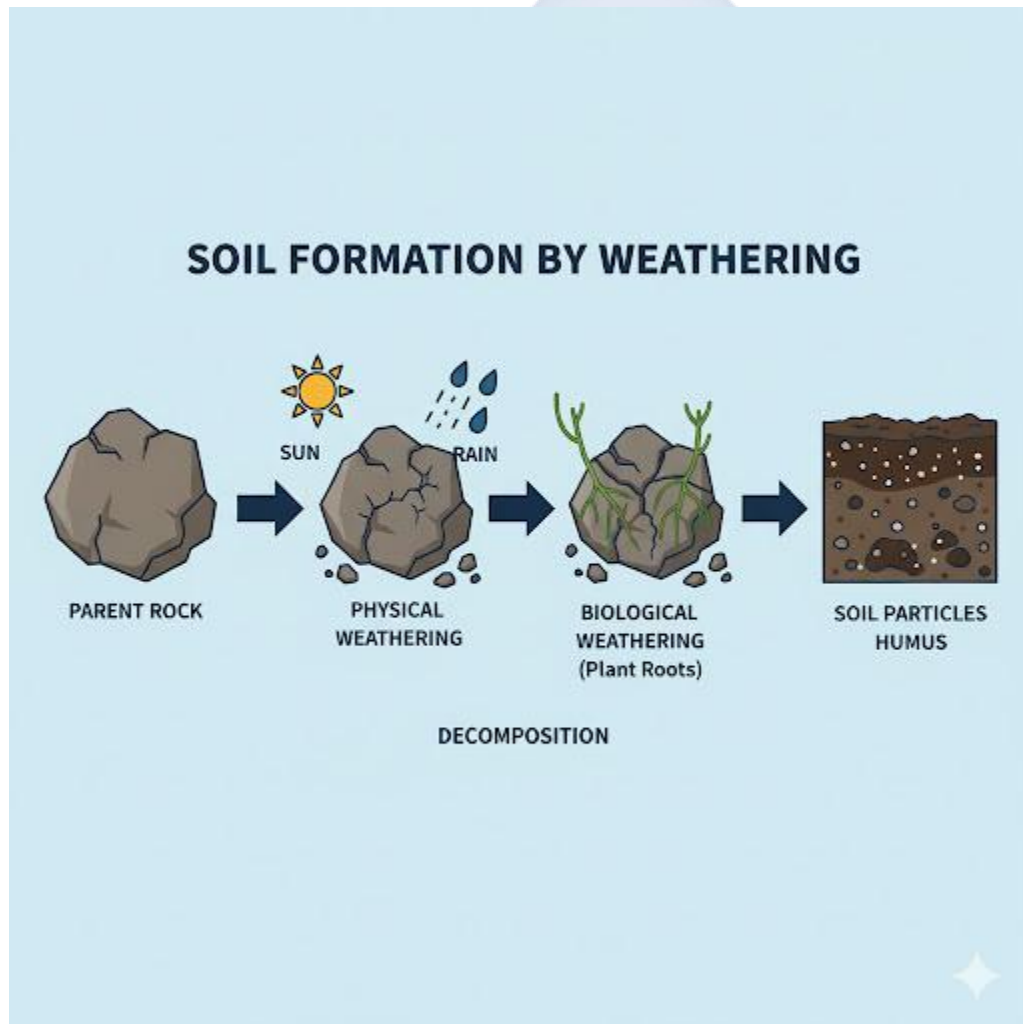


WEEK 3: SOIL FORMATION AND CONSERVATION

Topic: Soil Formation and Conservation

Breakdown (Subtopics):

- Meaning of soil and its formation (weathering)
- Types of soil and characteristics
- Soil erosion and conservation practices
- Importance of soil to living things



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Soil" and "Weathering."

2. Explain how soil is formed over time.
3. List the three main types of soil and one key characteristic of each.
4. Define "Soil Erosion" and list its two main agents.
5. List and explain three methods of soil conservation.

Lesson Content:

1. Meaning and Formation of Soil

- **Definition:** Soil is the **topmost layer of the Earth's crust**. It is a complex mixture of:
 1. Mineral particles (from weathered rock)
 2. Organic matter (dead and decaying plants/animals, called **humus**)
 3. Air
 4. Water
- **Formation (Weathering):** Soil is formed over thousands or millions of years by a slow process called **weathering**.
 - **Weathering** is the *breaking down* of large rocks into smaller and smaller pieces.
 - **Types of Weathering:**
 - **Physical Weathering:** Breaking rocks without changing their chemical composition (e.g., changes in temperature—heating and cooling, frost action—water freezing in cracks, plant roots growing into rocks).
 - **Chemical Weathering:** Breaking rocks by chemical reactions (e.g., acid rain dissolving limestone).
 - These small rock particles then mix with humus (from decomposers) to form fertile soil.

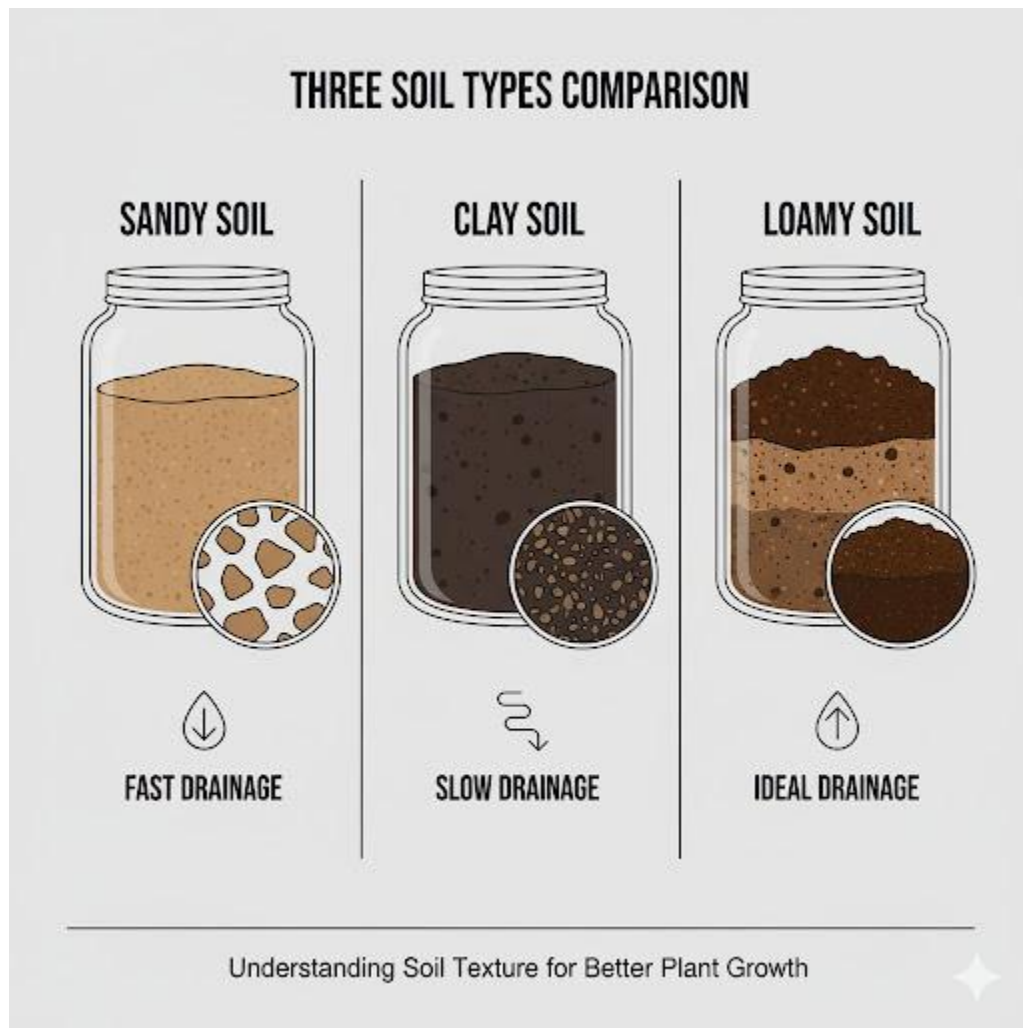
2. Types of Soil and Characteristics

Soils are classified by the size of their particles.

- **A. Sandy Soil**
 - **Particles:** Large and coarse (you can feel the grains).
 - **Characteristics:** Feels gritty. Has large air spaces. **Water drains through it very quickly** (low water retention). Dries out fast. Not fertile (low in humus).

- **B. Clay Soil**

- **Particles:** Very small and fine.
- **Characteristics:** Feels smooth and sticky when wet, gets hard and cracks when dry. Tiny air spaces. **Holds water for a very long time** (high water retention). Can become waterlogged.



- **C. Loamy Soil (Loam)**

- **Particles:** A *mixture* of sand, clay, and silt.
- **Characteristics:** This is the **best soil for agriculture**. It holds enough water but also has enough air spaces for roots to breathe. It is usually dark and rich in humus.

3. Soil Erosion and Conservation

- **Soil Erosion:** This is the *washing or blowing away* of the fertile topsoil by the agents of erosion. This is a huge problem.
 - **Agents of Erosion:**
 1. **Water:** (Splash erosion, gully erosion) Heavy rain washes soil away.
 2. **Wind:** Strong winds blow dry, bare soil away.
- **Soil Conservation:** These are the methods used to *protect* the soil from being eroded and to maintain its fertility.
 1. **Afforestation/Cover Cropping:** Planting trees and cover crops (like grass or beans). The roots hold the soil together, and the leaves block the impact of rain and wind. (This is the best method).
 2. **Terracing:** Cutting flat, "step-like" platforms on a steep hill. This slows down the flow of water.
 3. **Contour Ploughing:** Ploughing *across* a slope, not up and down. The furrows act like small dams to trap water.
 4. **Crop Rotation:** Planting different crops on the same land in different seasons to return nutrients to the soil.

4. Importance of Soil

1. **Agriculture:** It is the medium for growing all our food (crops).
2. **Habitat:** It is a home for billions of organisms (earthworms, bacteria, fungi, insects).
3. **Anchorage:** It anchors plant roots so they can stand firm.
4. **Foundation:** We build our houses, roads, and schools on it.
5. **Water Filter:** It helps to filter and clean rainwater as it seeps underground.

Evaluation:

1. What is soil?
2. What is "weathering"?
3. List the three main types of soil. Which one is best for farming?
4. What is "soil erosion"? List its two main agents.
5. What is "afforestation," and how does it prevent erosion?

Assignment:

1. List the four main components of soil. What is "humus"?
2. What is the main difference between sandy soil and clay soil in terms of water retention?
3. What is "terracing," and where is it used?
4. List three reasons why soil is important to living things.
5. How can overgrazing (letting cows eat all the grass) lead to soil erosion?

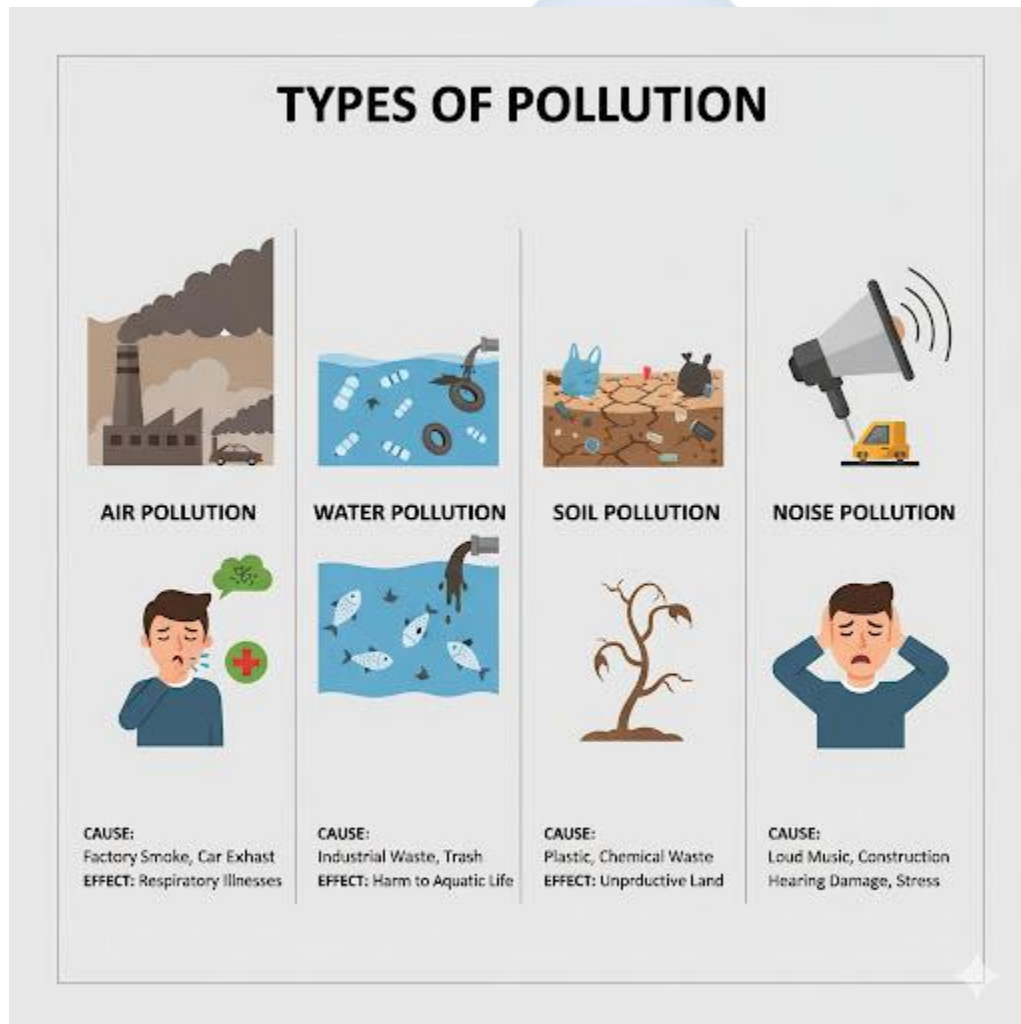


WEEK 4: ENVIRONMENTAL POLLUTION

Topic: Environmental Pollution

Breakdown (Subtopics):

- Meaning and types (air, water, soil, noise)
- Causes and effects
- Control and prevention methods
- Roles of individuals and government agencies



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Pollution" and "Pollutant."

2. List and define the four main types of pollution.
3. List a major cause and a major effect for each type of pollution.
4. Explain the general methods for controlling pollution.
5. Identify the main government agency in Nigeria responsible for environmental protection.

Lesson Content:

1. Meaning of Pollution

- **Pollution:** Is the introduction of **harmful substances or contaminants** (called **pollutants**) into the natural environment, causing harm, damage, or discomfort.
- **Pollutant:** The specific substance that causes the pollution (e.g., plastic waste, sewage, smoke, loud sound).

2. Types of Pollution: Causes and Effects

- **A. Air Pollution**
 - **Definition:** The release of harmful pollutants (gases, smoke, dust) into the atmosphere.
 - **Causes (Sources):**
 - Fumes from vehicle exhausts (cars, lorries, okada).
 - Smoke from factory chimneys and generators.
 - Burning of rubbish and bush burning.
 - Gases like Sulphur Dioxide (SO₂) and Carbon Monoxide (CO).
 - **Effects:**
 - **Health:** Respiratory (breathing) diseases like asthma, bronchitis, and lung cancer.
 - **Environment:** **Acid rain** (which kills trees and fish) and **Global Warming/Climate Change**.
- **B. Water Pollution**
 - **Definition:** The contamination of water bodies (rivers, lakes, oceans, groundwater).
 - **Causes (Sources):**

- **Sewage:** Dumping of untreated human waste (faeces, urine) into rivers.
- **Industrial Waste:** Factories dumping toxic chemicals (like mercury) into water.
- **Oil Spills:** (e.g., in the Niger Delta).
- **Agricultural Run-off:** Fertilizers and pesticides washed from farms into rivers.
- **Effects:**
 - **Health:** Spreads diseases like **cholera**, **typhoid**, and dysentery.
 - **Environment:** Kills fish and other aquatic life; destroys ecosystems.
- **C. Soil (Land) Pollution**
 - **Definition:** The contamination of land by solid or liquid waste.
 - **Causes (Sources):**
 - **Improper Waste Disposal:** Dumping refuse (especially non-biodegradable waste like **plastic bags/bottles**) on land.
 - **Agricultural:** Overuse of chemical fertilizers and pesticides.
 - **Industrial:** Chemical spills, illegal dumping of toxic waste.
 - **Effects:**
 - Makes soil infertile and unable to grow crops.
 - Poisons underground water.
 - Kills soil organisms (like earthworms).
- **D. Noise Pollution**
 - **Definition:** The presence of unwanted or disturbing sound that has a harmful effect on human and animal life.
 - **Causes (Sources):**
 - Loud generators.
 - Loud music from homes, cars, and religious centres (churches/mosques).
 - Hooting of horns in traffic.
 - Industrial machinery.
 - **Effects:**
 - **Health:** Hearing loss, deafness, high blood pressure, stress, and lack of sleep.

3. Control and Prevention of Pollution

1. **Legislation:** Government must make and *enforce* strong laws against pollution (e.g., fining factories that pollute).
2. **Waste Management:** Proper disposal of refuse (Week 7) and treatment of sewage.
3. **Public Education:** Teaching citizens (like in this class) about the dangers of pollution and the importance of the 3Rs (Reduce, Reuse, Recycle).
4. **Using Green Technology:** Shifting to cleaner energy (like solar power) instead of generators; using electric cars.



4. Role of Government Agencies

- **NESREA:** The main federal agency is the **National Environmental Standards and Regulations Enforcement Agency**. Their job is to set and enforce environmental standards (e.g., limits on smoke from factories).
- State-level agencies (like **LAWMA** in Lagos) manage waste.

Evaluation:

1. What is "pollution"?
2. What is a "pollutant"?
3. List the four main types of pollution.
4. What is a major cause of air pollution?
5. What is a major health effect of water pollution (e.g., a disease)?

Assignment:

1. Define Air, Water, Soil, and Noise pollution.
2. What is "acid rain"? Which type of pollution causes it?
3. What is the full meaning of NESREA?
4. Explain two ways "plastic" causes soil and water pollution.
5. List three things you can do *personally* to reduce pollution in your community.

WEEK 6: WASTE AND ITS MANAGEMENT

Topic: Waste and Its Management

Breakdown (Subtopics):

- Meaning and types of waste
- Waste disposal methods
- Reduce, Reuse, Recycle (3Rs)
- Effects of poor waste management



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Waste" (or "Refuse").

2. List and differentiate the three main types of waste by state (solid, liquid, gaseous).
3. Define "Biodegradable" and "Non-biodegradable" waste and give examples.
4. Explain the 3Rs (Reduce, Reuse, Recycle) in detail.
5. List three major effects of poor waste management.

Lesson Content:

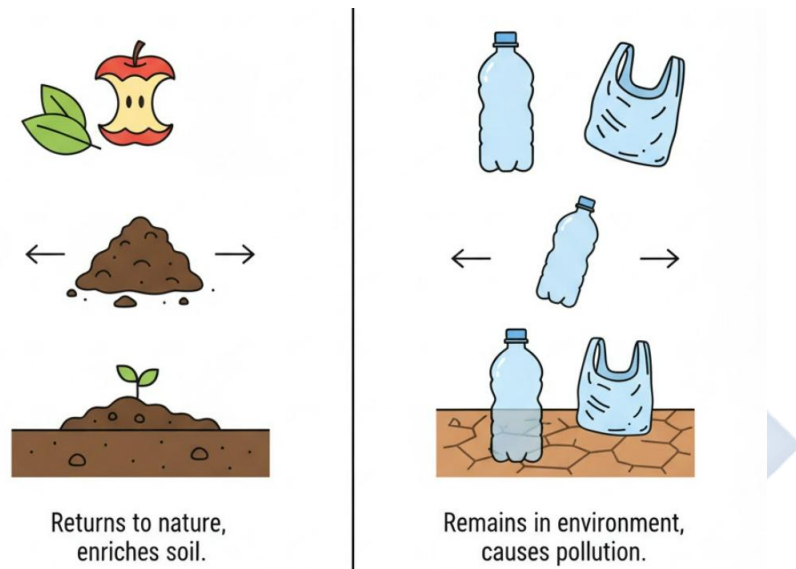
1. Meaning and Types of Waste

- **Definition: Waste** (also called refuse, rubbish, or garbage) is any material that is unwanted, has been used, and is discarded (thrown away).
- **Types of Waste (by State):**
 1. **Solid Waste:** All solid or semi-solid refuse (e.g., paper, food scraps, plastic bottles, metal cans).
 2. **Liquid Waste:** All waste in liquid form (e.g., **sewage** (from toilets), dirty water (from bathing/washing), industrial chemical waste).
 3. **Gaseous Waste:** All waste in gas form (e.g., smoke from generators, vehicle exhaust, factory fumes).
- **Types of Waste (by Decomposition):**
 - **Biodegradable:** Waste that can be *broken down* (decomposed) by bacteria and fungi. It rots.
 - *Examples:* Food scraps, paper, leaves, grass, cotton, human waste.
 - **Non-Biodegradable:** Waste that *cannot* be broken down by micro-organisms. It lasts for hundreds of years.
 - *Examples:* **Plastic** (bottles, bags), glass, metal, nylon. (This is the most dangerous type of waste).

2. Waste Disposal Methods

- **A. Poor / Improper Methods (These are bad):**
 - **Open Dumping:** Throwing refuse on the roadside, in gutters, or in any open space.

- **Dumping in Water:** Throwing waste directly into rivers or streams.
- **B. Proper / Modern Methods:**
 - **Sanitary Landfill:** A large, prepared site where waste is buried safely in layers, covered with soil to prevent smells and pests.
 - **Incineration:** Burning waste at very high temperatures in a controlled factory (incinerator) to reduce its volume. Can be used to generate electricity.
 - **Recycling:**



3. The 3Rs: The Best Waste Management Strategy This is the "Waste Hierarchy." The most important is Reduce.

1. REDUCE:

- **Meaning:** To *use less* stuff. This is the best way to manage waste. If you don't create waste, you don't have to manage it.
- **Examples:** Use a reusable water bottle (instead of buying sachet water), write on both sides of a paper, buy things with less packaging.

2. REUSE:

- **Meaning:** To *use an item again* for the same or a new purpose.
- **Examples:** Use an old ice cream container to store food, donate old clothes, repair a broken chair instead of throwing it away.

3. RECYCLE:

- **Meaning:** To *convert* (process) waste materials into new, useful products.
- **Examples:** Melting plastic bottles to make new chairs, melting glass bottles to make new glass, pulping old newspapers to make new paper.

4. Effects of Poor Waste Management (This links to Week 3 and Week 4)

1. **Disease Outbreak:** Piles of refuse attract rats, flies, and mosquitoes, which spread diseases like Cholera, Lassa fever, and Malaria.
2. **Flooding:** When refuse is dumped in gutters and drains, it blocks water, causing severe flooding when it rains.
3. **Pollution:** Waste pollutes the air (bad smells, burning), water (sewage), and soil (plastics, chemicals).
4. **Ugliness:** It makes the community look dirty, ugly, and unattractive.

Evaluation:

1. What is "waste"?
2. What is the difference between solid waste and liquid waste?
3. What is the difference between biodegradable and non-biodegradable waste?
4. List the 3Rs in order of importance.
5. Explain two major effects of poor waste management on a community.

Assignment:

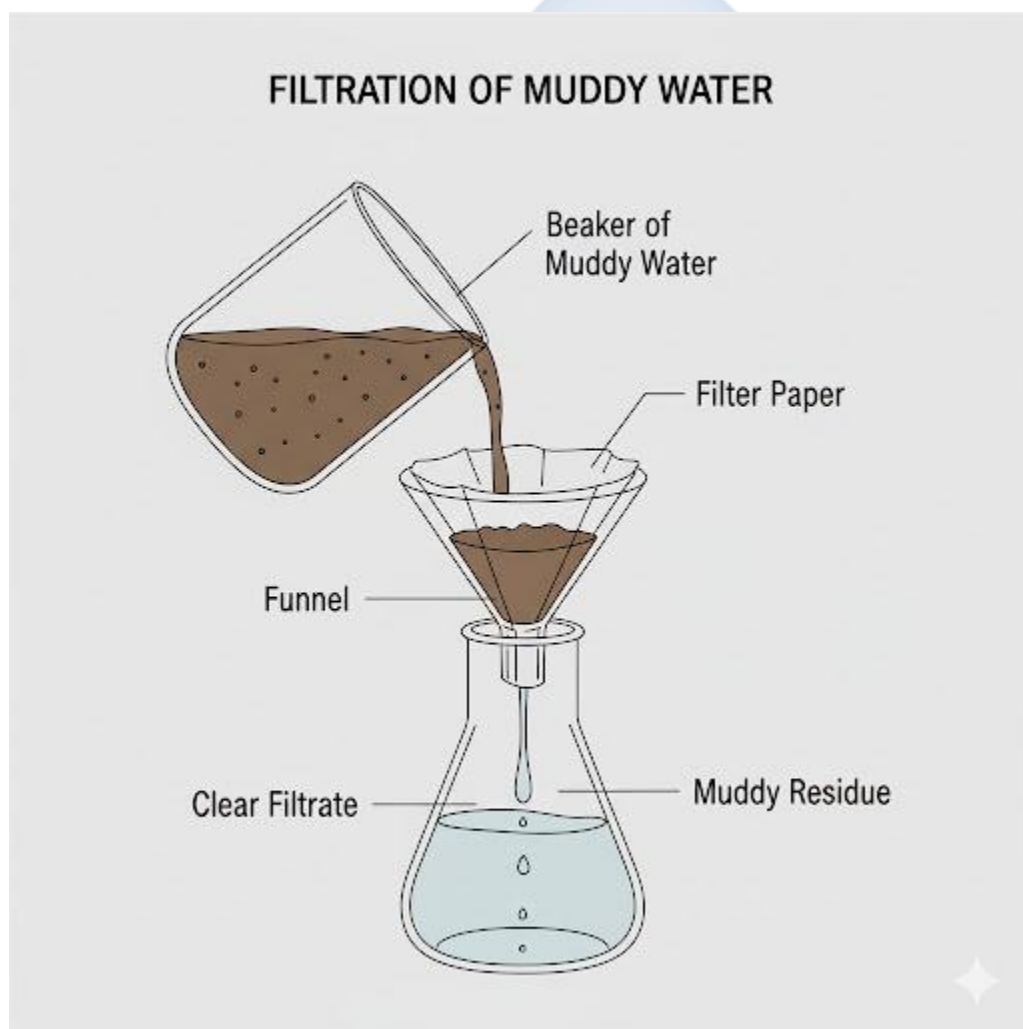
1. Give two examples of biodegradable waste and two examples of non-biodegradable waste.
2. What is a "sanitary landfill"?
3. Explain the difference between "Reuse" and "Recycle" using a plastic bottle as an example. (Reuse = refilling it with water. Recycle = melting it).
4. How does improper sewage disposal (a liquid waste) lead to disease?
5. What is the main solid waste you produce in your home, and how can you *Reduce* it?

WEEK 8: AIR AND WATER POLLUTION

Topic: Air and Water Pollution

Breakdown (Subtopics):

- Sources of air and water pollution
- Effects on living organisms
- Prevention and purification methods
- Experiment: Filtration of muddy water



Performance Objectives: By the end of this lesson, students should be able to:

1. List three specific sources and three specific effects of air pollution.

2. List three specific sources and three specific effects of water pollution.
3. Describe the process of water purification (filtration, boiling).
4. Perform a simple experiment to filter muddy water.

Lesson Content:

1. Air Pollution (Deep Dive)

- **Specific Sources (Pollutants):**

1. **Carbon Monoxide (CO):** A highly poisonous, invisible, odourless gas. Comes from *incomplete combustion* in car/lorry exhaust and "I-pass-my-neighbour" generators.
2. **Sulphur Dioxide (SO₂):** Comes from burning fossil fuels (like coal and oil) in power plants and factories.
3. **Particulate Matter (Soot):** Tiny black particles of carbon from diesel engines and burning rubbish.
4. **Greenhouse Gases (CO₂):** Carbon Dioxide from all forms of burning.

- **Specific Effects:**

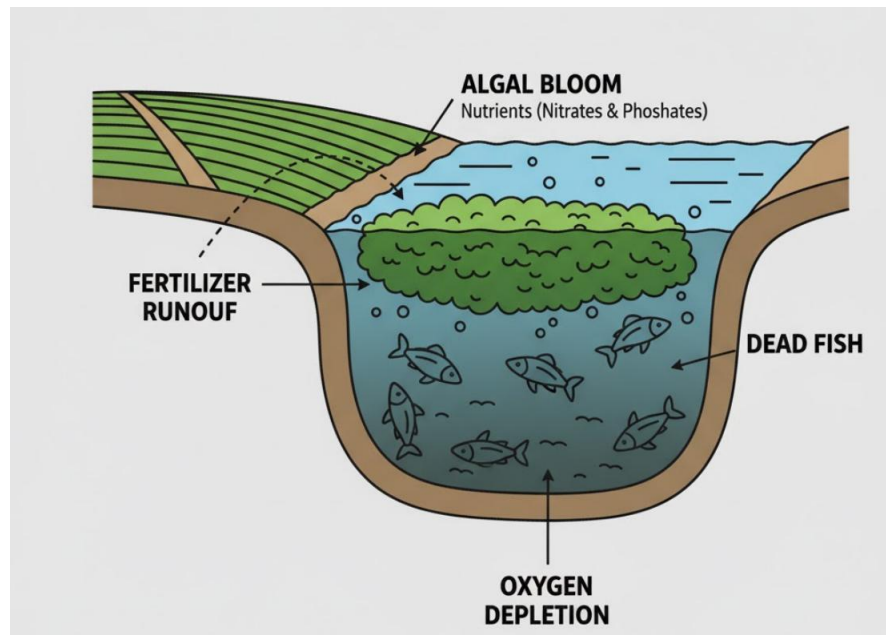
1. **Health:** Carbon Monoxide reduces oxygen in the blood and can kill. Soot and SO₂ cause asthma, bronchitis, and lung cancer.
2. **Acid Rain:** SO₂ reacts with water in the atmosphere to form sulphuric acid. This acid rain falls and kills trees, poisons lakes (killing fish), and damages buildings.
3. **Global Warming:** (See Week 10). CO₂ traps heat in the atmosphere.

2. Water Pollution

- **Specific Sources (Pollutants):**

1. **Sewage and Domestic Waste:** This is the main source. It contains human faeces and urine, which are full of *pathogens* (disease-causing bacteria and viruses).
2. **Industrial Waste:** Factories dump *toxic chemicals* like mercury, lead, and acid directly into rivers.
3. **Oil Spillage:** Crude oil leaks from pipelines (e.g., in Niger Delta), covering the water, killing fish, and destroying mangrove swamps.

4. **Agricultural Run-off:** Rain washes excess fertilizers and pesticides from farms into rivers.



- **Specific Effects:**

1. **Disease:** Spreads water-borne diseases like **Cholera, Typhoid, and Dysentery**.
2. **Death of Aquatic Life:** Oil spills prevent oxygen from entering the water, suffocating fish. Toxic chemicals poison them directly.
3. **Eutrophication:** Fertilizers from farms cause a massive "bloom" of algae (green plants) on the water surface. When this algae dies, decomposers use up *all* the oxygen in the water, creating a "dead zone" where no fish can survive.

3. Prevention and Purification

- **Prevention:**

- Treat sewage before releasing it into rivers.
- Enforce laws (NESREA) to stop factories from dumping chemicals.
- Maintain pipelines to prevent oil spills.

- **Purification (How to make water safe to drink):**

1. **Filtration:** Passing water through a filter (like filter paper or a special cloth) to remove *solid* particles (like mud, sand, and dirt).
2. **Boiling:** Heating water to 100°C for several minutes. This *kills* the germs (bacteria and viruses).
3. **Adding Chemicals (Chlorination):** Adding a small, safe amount of chlorine to kill germs (this is done by city water corporations).

4. Experiment: Filtration of Muddy Water

- **Aim:** To separate mud (an insoluble solid) from water.
- **Apparatus:** Beaker, muddy water, funnel, filter paper, conical flask, tripod stand.
- **Method:**
 1. Fold the filter paper into a cone shape and place it in the funnel.
 2. Place the funnel over the mouth of the conical flask.
 3. Slowly pour the muddy water into the filter paper.
- **Observation:**
 - The clear, clean water (the **filtrate**) passes through the tiny holes in the paper and collects in the flask.
 - The mud and sand particles (the **residue**) are too big and are trapped in the filter paper.
- **Conclusion:** Filtration can be used to separate an insoluble solid from a liquid. (Note: This does *not* kill germs).

Evaluation:

1. What poisonous gas is produced by generators and car exhaust?
2. What is "acid rain," and what causes it?
3. What is "eutrophication"?
4. What is the difference between "filtration" and "boiling" when purifying water?
5. In the filtration experiment, what is the "residue" and what is the "filtrate"?

Assignment:

1. List two sources and two effects of air pollution.
2. List two sources and two effects of water pollution.
3. What is the danger of oil spills in the Niger Delta?
4. Why is filtration *not* enough to make river water safe for drinking? What else must you do?
5. Draw and label the apparatus for the "Filtration of Muddy Water" experiment.



WEEK 9: CONSERVATION OF NATURAL RESOURCES

Topic: Conservation of Natural Resources

Breakdown (Subtopics):

- Meaning and types of natural resources
- Renewable vs. non-renewable resources
- Importance of conservation
- Role of individuals in conservation

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Natural Resources."
2. Define "Renewable Resources" and give 3 examples.
3. Define "Non-renewable Resources" and give 3 examples.
4. Define "Conservation."
5. Explain three reasons why conservation is important.

Lesson Content:

1. What are Natural Resources? **Natural Resources** are materials or substances that exist in **nature** (they are not man-made) and can be used by humans to meet their needs and create wealth.

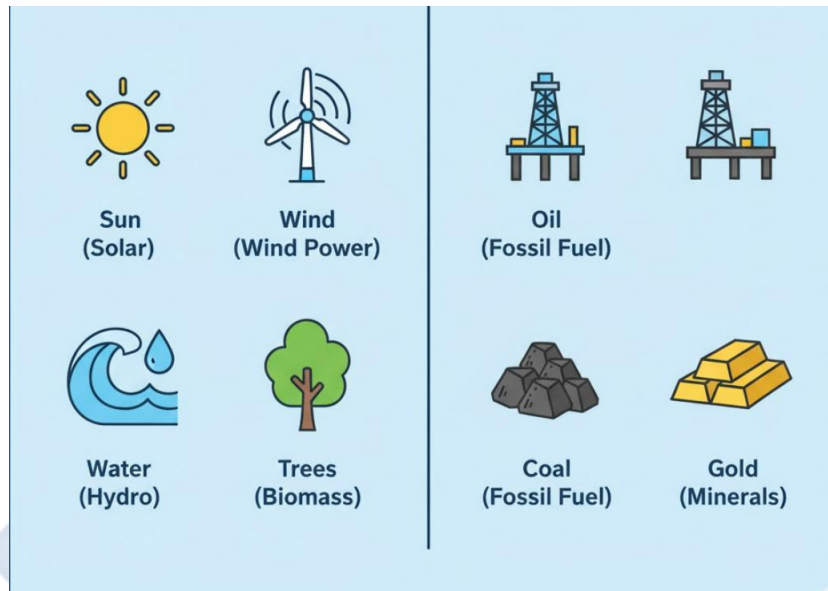
- *Examples:* Soil, water, air, sunlight, forests (wood), animals, crude oil, iron, gold.

2. Types of Natural Resources This is the most important classification.

A. Renewable Resources

- **Definition:** These are resources that **can be replaced or regenerated** naturally over a relatively short period. They will *not* run out if we manage them wisely.
- **Examples:**
 - **Forests (Trees):** We can plant new trees (afforestation).

- **Water:** It is replaced by rain (the water cycle).
- **Sunlight (Solar Energy):** Will not run out for billions of years.
- **Wind:** Will not run out.
- **Animals/Fish:** They can reproduce and make more.
- **Soil:** Can be made fertile again.



B. Non-Renewable Resources

- **Definition:** These are resources that **cannot be replaced** once they are used up. They were formed over millions of years and their supply is *finite* (limited).
- **Examples:**
 - **Fossil Fuels:** (Formed from dead, ancient organisms)
 - **Crude Oil (Petroleum)**
 - **Natural Gas**
 - **Coal**
 - **Minerals:**
 - **Gold, Silver, Diamonds**
 - **Iron ore, Bauxite (for aluminium)**

3. What is Conservation? Conservation is the **wise, careful, and responsible use and management** of natural resources to prevent their waste, destruction, or depletion (finishing).

- It does *not* mean "don't use." It means "use *wisely* so that it lasts."

4. Importance of Conservation (Why must we conserve?)

1. **To Prevent Depletion:** To ensure that non-renewable resources (like oil) do not finish, and that renewable resources (like fish) are not used up faster than they can be replaced.
2. **For Future Generations:** To ensure that our children and grandchildren will also have resources to use. This is called **sustainability**.
3. **To Maintain Ecological Balance:** To protect ecosystems. If we cut down all the forests (deforestation), we destroy animal habitats, increase soil erosion, and worsen climate change.
4. **To Save Money:** Using less energy (e.g., turning off lights) saves money and conserves resources.

5. Role of Individuals in Conservation

1. **Practice the 3Rs: Reduce, Reuse, Recycle.** This conserves resources (minerals, trees, oil) used to make new products.
2. **Conserve Energy:** Turn off lights and appliances when not in use. Walk or use a bicycle for short trips instead of a car.
3. **Conserve Water:** Close taps when not in use; repair leaks.
4. **Afforestation:** Plant trees, do not cut them down unnecessarily.
5. **Use Renewable Energy:** Use solar panels instead of generators (which burn non-renewable fuel).

Evaluation:

1. What are "natural resources"?
2. What is the main difference between a renewable and a non-renewable resource?
3. Is crude oil renewable or non-renewable? Why?

4. Is a forest (trees) renewable or non-renewable? Why?
5. What is "conservation"?

Assignment:

1. Define Renewable and Non-renewable resources.
2. List three examples of each type.
3. What is "sustainability"? (Linked to conservation).
4. List three major reasons why conservation is important.
5. List four actions *you* can take as an individual to conserve natural resources.

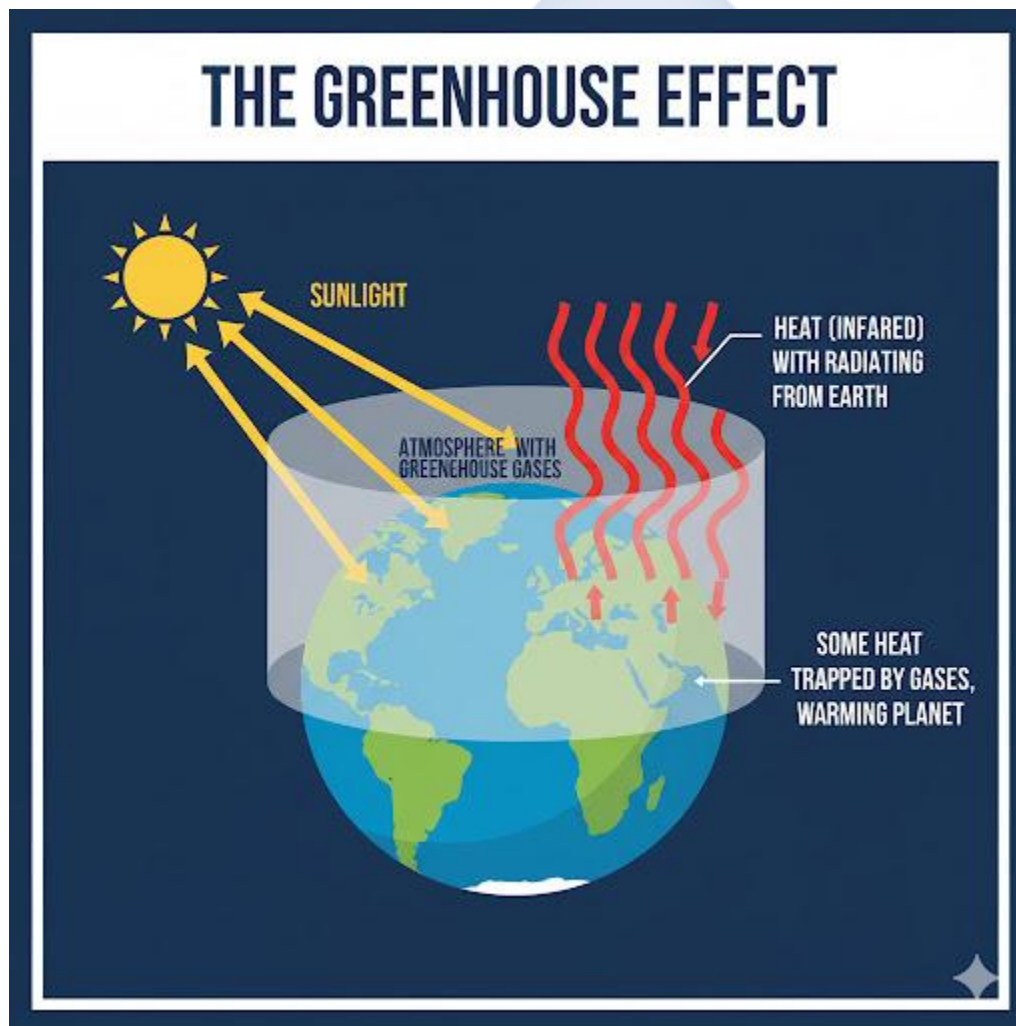


WEEK 10: CLIMATE AND ENVIRONMENTAL CHANGES

Topic: Climate and Environmental Changes

Breakdown (Subtopics):

- Meaning and causes of climate change
- Effects of global warming
- Mitigation and adaptation strategies
- School-based tree planting projects



Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between "Weather" and "Climate."

2. Define "Climate Change" and "Global Warming."
3. Explain the "Greenhouse Effect" as the main cause of global warming.
4. List three major effects of climate change.
5. Differentiate between "mitigation" and "adaptation" strategies.

Lesson Content:

1. Weather vs. Climate

- **Weather:** This is the short-term condition of the atmosphere in a *specific place* at a *specific time*.
 - *Example:* "It is rainy and cold in Lagos *today*."
- **Climate:** This is the *average* weather pattern of a region over a *long period* (e.g., 30 years or more).
 - *Example:* "Lagos has a *tropical climate* with a wet season and a dry season."

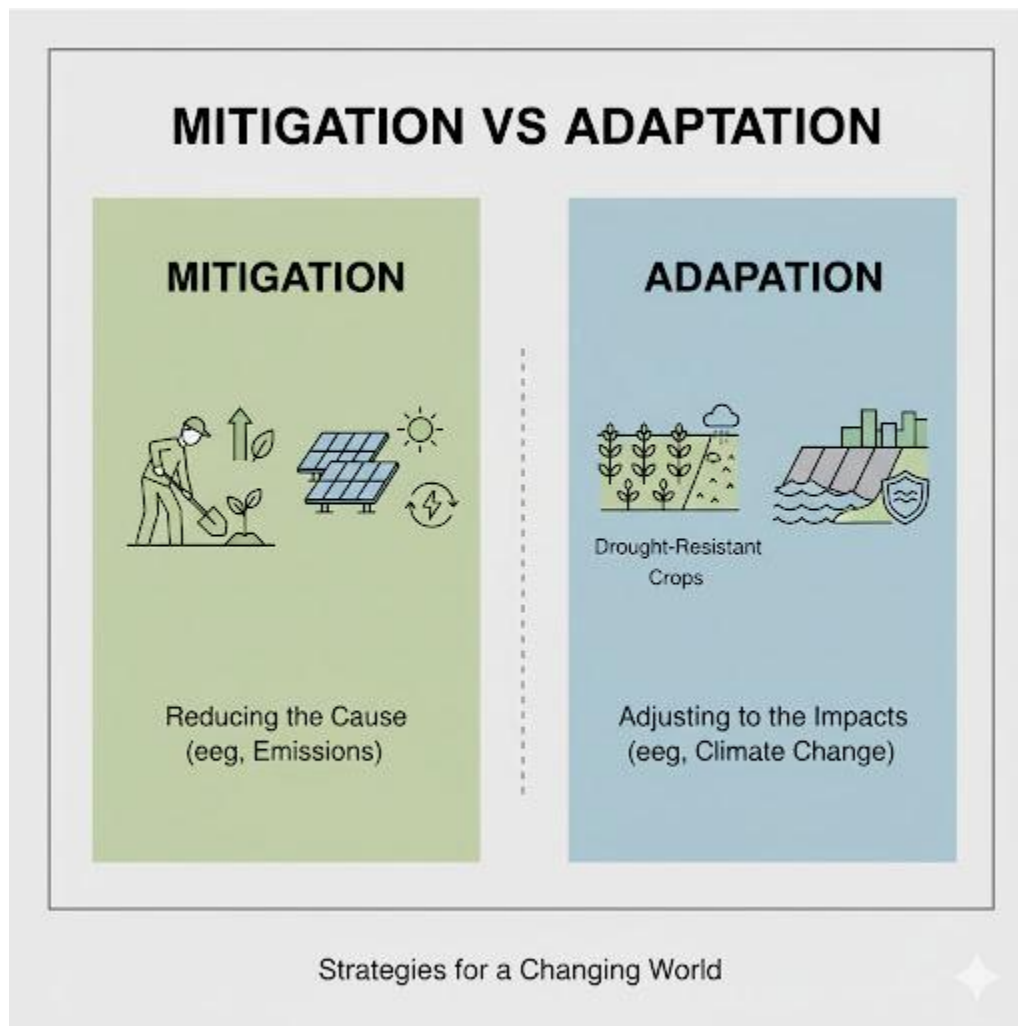
2. Climate Change and Global Warming

- **Climate Change:** This is a significant, long-term change in the average weather patterns of the Earth.
- **Global Warming:** This is a key part of climate change. It is the long-term *heating up* of the Earth's average surface temperature.

3. Causes of Climate Change (The Greenhouse Effect)

1. The Earth is surrounded by a layer of gases called the **atmosphere**.
2. Sunlight (solar energy) passes through the atmosphere and warms the Earth.
3. The Earth radiates some of this heat back towards space.
4. A group of gases called **Greenhouse Gases** (GHGs) in the atmosphere (especially **Carbon Dioxide, CO₂**) act like a blanket, *trapping* this heat and keeping the Earth warm. This is normal and good (it's called the **Greenhouse Effect**).

5. **The Problem:** Human activities, especially **burning fossil fuels** (oil, gas, coal for cars, generators, factories) and **deforestation** (cutting down trees, which absorb CO₂), have released *too much* CO₂ into the atmosphere.
6. This *extra* CO₂ has made the "blanket" thicker, trapping *too much* heat and causing the Earth to get warmer (Global Warming).



4. Effects of Climate Change / Global Warming

1. **Melting Ice Caps:** Glaciers and ice at the North and South Poles are melting.
2. **Rising Sea Levels:** The melted ice flows into the oceans, causing sea levels to rise. This will flood coastal cities (like Lagos).
3. **Extreme Weather:** More frequent and intense heatwaves, droughts (no rain), and floods (too much rain).

4. **Disruption to Agriculture:** Droughts and floods destroy farms, leading to food shortages.
5. **Desertification:** Good land turning into desert (a major problem in Northern Nigeria).

5. Solutions: Mitigation and Adaptation

- **A. Mitigation (Tackling the Cause):**

- **Meaning:** Actions taken to *reduce* the amount of greenhouse gases we release.
- **Examples:**
 - **Afforestation: Planting trees** (they absorb CO₂).
 - **Use Renewable Energy:** Using solar, wind, and hydro power instead of generators.
 - **Energy Efficiency:** Using less energy (turning off lights, buying fuel-efficient cars).

- **B. Adaptation (Dealing with the Effects):**

- **Meaning:** Actions taken to *adjust* to the changes that are already happening.
- **Examples:**
 - Building sea walls to protect coastal cities from rising sea levels.
 - Farmers planting crops that are resistant to drought.
 - Building better drainage systems to cope with floods.

6. School-Based Project:

- A simple *mitigation* strategy is to organize a "Tree Planting Day" in the school compound to help absorb CO₂.

Evaluation:

1. What is the difference between weather and climate?
2. What is "global warming"?
3. What is the main human activity that causes global warming?
4. List three major effects of climate change.
5. What is the difference between "mitigation" and "adaptation"?

Assignment:

1. Define "Climate Change" and "Greenhouse Effect."
2. What is the most important greenhouse gas, and where does it come from?
3. Give one example of a *mitigation* strategy.
4. Give one example of an *adaptation* strategy.
5. How would *rising sea levels* specifically affect a city like Lagos?



Week 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Comprehensive review of all topics in Term 1

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and define the key concepts of the term.
2. Explain the connections between the environment, pollution, and conservation.
3. Discuss the role of citizens in managing waste and protecting resources.

Lesson Content:

1. Summary of Term 1 Topics (The "Environment" Term):

- **Week 1 (Review):** Matter, Energy, Lab Safety, Observation, Measurement.
- **Week 2 (Ecosystems):** Biotic (Producers, Consumers, Decomposers) and Abiotic (Sun, Air, Water, Soil) components are all interdependent. Energy flows in Food Chains (e.g., Grass -> Goat -> Lion) and Food Webs.
- **Week 3 (Soil):** Formed by weathering of rocks. Types: Sandy (drains fast), Clay (holds water), Loamy (best). Threatened by Erosion (by wind/water). Protected by Conservation (e.g., planting trees - afforestation, terracing).
- **Week 4 (Pollution):** Introducing pollutants (harmful substances). Types: Air (smoke, CO₂), Water (sewage, oil), Soil (plastic), Noise (generators). Agency: NESREA.
- **Week 7 (Waste):** Biodegradable (rots) vs. Non-biodegradable (plastic). Management: 3Rs (Reduce, Reuse, Recycle). Poor management causes flooding and disease.
- **Week 8 (Pollution Deep Dive):** Air pollution causes acid rain and health issues. Water pollution (sewage, fertilizers) causes cholera, typhoid, and eutrophication.
- **Week 9 (Conservation):** Wise use of resources. Renewable (can be replaced - trees, sun) vs. Non-renewable (will finish - oil, gas).

- **Week 10 (Climate Change):** Long-term weather change caused by Global Warming. Caused by the Greenhouse Effect (too much CO₂ from burning fuel, deforestation). Effects: Rising sea levels, floods, drought. Solutions: Mitigation (planting trees) and Adaptation (building sea walls).

2. Class Activity: Connecting the Dots (The Big Picture)

- How does Deforestation (cutting trees) link to:
 - Soil Erosion (Week 3)? (No roots to hold soil).
 - Climate Change (Week 10)? (Fewer trees to absorb CO₂).
 - Loss of Habitat (Week 2)? (Animals lose their homes).
- How does Poor Waste Management (Week 7) link to:
 - Water Pollution (Week 8)? (Refuse and sewage enter rivers).
 - Soil Pollution (Week 4)? (Plastics on land).
 - Flooding (Week 3/7/10)? (Refuse blocks drains).
- How does Burning Fossil Fuels (Week 9) link to:
 - Air Pollution (Week 8)? (Releases CO, SO₂).
 - Climate Change (Week 10)? (Releases CO₂).

Evaluation: This week's evaluation is the First Term Examination.

Assignment:

1. Study all notes from Week 1 to Week 10 for the examination.
2. Draw a simple food web with at least 5 organisms (including a producer, a primary consumer, a secondary consumer, and a decomposer).
3. Explain the difference between Mitigation and Adaptation as solutions to climate change.

SECOND TERM – JSS 2 INTERMEDIATE SCIENCE LESSON NOTES

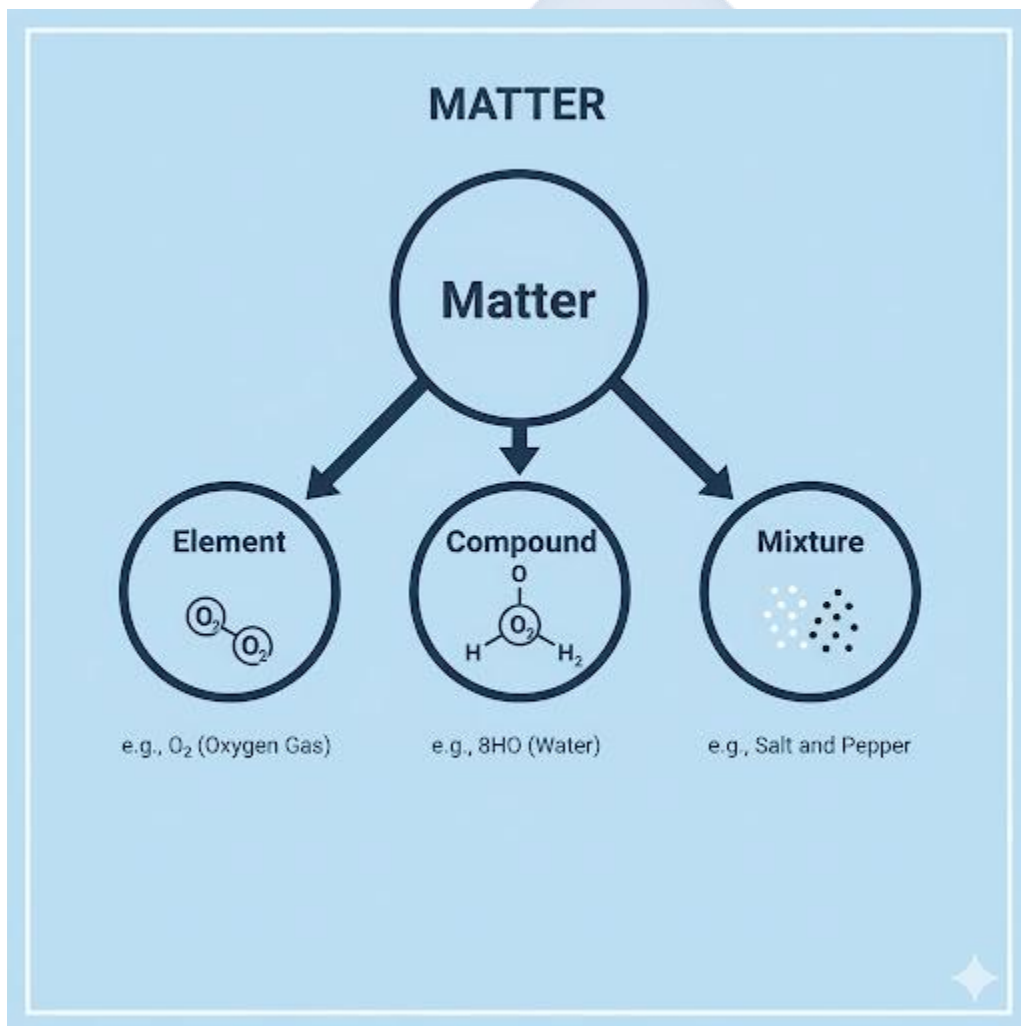


WEEK 1: ELEMENTS, COMPOUNDS, AND MIXTURES

Topic: Elements, Compounds, and Mixtures

Breakdown (Subtopics):

- Meaning and examples of elements, compounds, and mixtures
- Differences between elements, compounds, and mixtures
- Simple separation techniques
- Practical demonstrations



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Element," "Compound," and "Mixture" with examples.

2. List the key differences between a compound and a mixture in a table.
3. Define "Atom" as the smallest part of an element and "Molecule" as the smallest part of a compound.
4. List and describe four simple separation techniques.
5. State the correct separation technique for a given simple mixture.

Lesson Content:

1. Definitions All matter (Week 1, Term 1) can be classified as elements, compounds, or mixtures.

- **A. Element:**

- **Definition:** An element is a **pure substance** that cannot be broken down into anything simpler by any chemical means. It is made of only **one type of atom**.
- **Atom:** The smallest particle of an element that can exist and still have the properties of that element.
- **Examples:**
 - Metals: Iron (Fe), Gold (Au), Copper (Cu), Sodium (Na)
 - Non-Metals: Oxygen (O), Hydrogen (H), Carbon (C), Sulphur (S)

- **B. Compound:**

- **Definition:** A compound is a **pure substance** formed when **two or more different elements are chemically combined** in a **fixed ratio**.
- **Key Properties:**

1. The elements lose their original properties and form a *new substance* with *new properties*. (e.g., H and O are gases, but Water (H₂O) is a liquid).
2. They can only be separated by a *chemical reaction* (e.g., electrolysis).
 - **Molecule:** The smallest particle of a compound that can exist and still have the properties of that compound.
 - **Examples:** Water (H₂O), Carbon Dioxide (CO₂), Sodium Chloride (Table Salt, NaCl), Rust (Iron Oxide, Fe₂O₃).

- **C. Mixture:**

- **Definition:** A mixture consists of **two or more substances (elements or compounds) that are physically mixed** together but **not chemically combined**.
- **Key Properties:**
 0. The substances are *not* in a fixed ratio.
 1. Each substance *keeps its original properties*. (e.g., in a salt-water mixture, the water is still wet and the salt still tastes salty).
 2. They can be separated by *physical means*.
 - **Examples:** Air (a mixture of gases like Nitrogen, Oxygen), Salt water, Sand and water, Rice and stones, Fruit salad.

2. Differences between a Compound and a Mixture

Feature	Compound	Mixture
Formation	Formed by a chemical reaction.	Formed by a physical mixing.
Ratio	Elements are combined in a fixed ratio (e.g., H ₂ O is <i>always</i> 2:1).	Substances can be mixed in <i>any</i> ratio.
Properties	A new substance is formed with new properties .	Each substance keeps its original properties.
Separation	Can <i>only</i> be separated by chemical methods.	Can be separated by physical methods.
Energy	Energy (heat/light) is usually given out or absorbed during formation.	Little or no energy is involved in mixing.

3. Simple Separation Techniques (for Mixtures)

1. Filtration:

- **Use:** To separate an **insoluble solid** (does not dissolve) from a **liquid**.
- **Example:** Separating **sand from water**.
- **Method:** The mixture is poured through filter paper. The liquid (filtrate) passes through, and the solid (residue) is trapped.

2. Evaporation:

- **Use:** To separate a **soluble solid** (a *solute*, like salt) from a **liquid** (a *solvent*, like water).
- **Example:** Separating **salt from salt water**.
- **Method:** The solution is heated. The water (solvent) evaporates as steam, leaving the solid salt (solute) behind.

3. Sieving (or Sifting):

- **Use:** To separate **solid particles of different sizes**.
- **Example:** Separating **rice from stones**, or **flour from weevils**.
- **Method:** The mixture is passed through a sieve (a mesh). The smaller particles pass through, and the larger particles are left behind.

4. Magnetic Separation:

- **Use:** To separate a **magnetic substance** from a **non-magnetic** one.
- **Example:** Separating **iron filings** from **sand** or **sulphur**.
- **Method:** A magnet is passed over the mixture. It attracts the iron filings, leaving the other substance behind.

Evaluation:

1. What is the difference between an element and a compound?
2. What is the difference between a compound and a mixture?
3. What is the smallest particle of an element called?
4. What separation technique would you use to separate salt from water?
5. What separation technique would you use to separate sand from water?

Assignment:

1. Define Element, Compound, and Mixture.
2. Give two examples of each.
3. Create a table with two columns showing four key differences between a compound and a mixture.
4. What separation technique would you use to separate iron filings from rice?

5. What separation technique would you use to separate "gari" (cassava flour) from small stones?

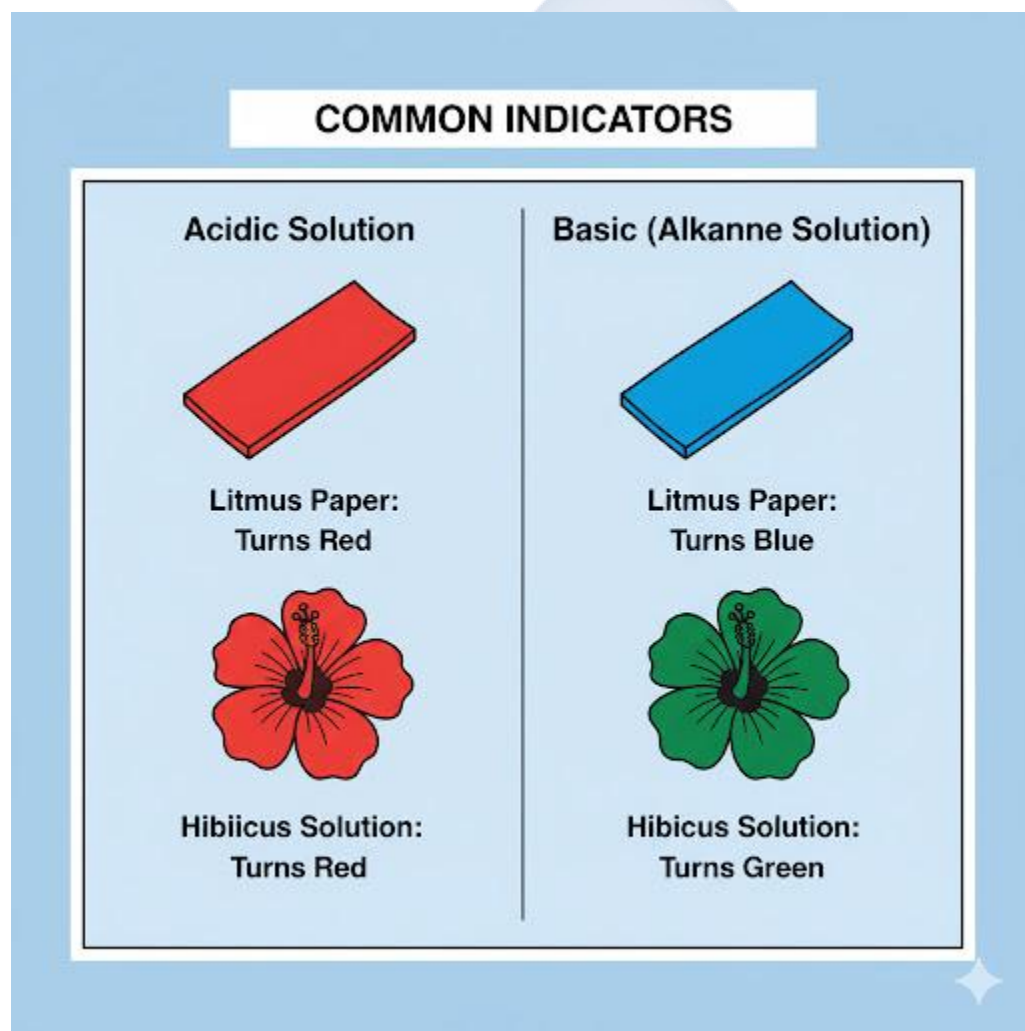


WEEK 2: ACIDS, BASES, AND SALTS

Topic: Acids, Bases, and Salts

Breakdown (Subtopics):

- Meaning and identification of acids and bases
- Common household acids and bases
- Indicators (litmus, turmeric, methyl orange)
- Neutralization reactions



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Acid" and list three of its properties.

2. Define "Base" and "Alkali" and list three of their properties.
3. Define "Indicator" and state the colour changes of litmus.
4. Identify common household substances as acidic or basic.
5. Define "Neutralization" and state its products.

Lesson Content:

1. Acids

- **Definition:** An acid is a chemical substance that has a **sour taste**, is corrosive (can "eat" through metal or cloth), and **turns blue litmus paper red**.
- **Properties:**
 1. They have a sour taste (Warning: NEVER taste chemicals in a lab).
 2. They turn blue litmus paper red.
 3. They are corrosive.
 4. They react with bases to form salt and water.
- **Common Acids:**
 - **Household Acids (Weak):**
 - Lime/Lemon/Orange juice (contains **Citric Acid**)
 - Vinegar (contains **Acetic Acid**)
 - Yoghurt (contains **Lactic Acid**)
 - **Laboratory Acids (Strong/Mineral):**
 - Hydrochloric Acid (HCl)
 - Sulphuric Acid (H₂SO₄)
 - Nitric Acid (HNO₃)

2. Bases

- **Definition:** A base is a chemical substance that has a **bitter taste**, feels **soapy** to the touch, and **turns red litmus paper blue**.
- **Alkali:** A base that can **dissolve in water** is called an alkali (e.g., caustic soda). All alkalis are bases, but not all bases are alkalis.

- **Properties:**

1. They have a bitter taste.
2. They feel soapy or slippery.
3. They turn red litmus paper blue.
4. They are corrosive (strong bases like caustic soda).
5. They react with acids to form salt and water.

- **Common Bases (Alkalis):**

- **Household Bases:**

- Soap / Detergent
 - Baking Soda (Sodium Bicarbonate)
 - Wood Ash (contains Potassium Hydroxide)
 - Toothpaste

- **Laboratory Bases:**

- Sodium Hydroxide (NaOH - "Caustic Soda")
 - Calcium Hydroxide (Lime water)

3. Indicators

- **Definition:** An indicator is a special chemical (dye) that **changes colour** when it is placed in an acidic or a basic solution. It "indicates" if a substance is an acid or a base.
- **Common Indicators and their Colour Changes:**

Indicator	Colour in Acid	Colour in Base	Colour in Neutral
Litmus Paper	Red	Blue	No change
Methyl Orange	Red	Yellow	Orange
Turmeric (Local)	Yellow	Reddish-brown	Yellow
Hibiscus (Zobo) Flower	Red/Pink	Green	Purple

4. Neutralization Reactions

- **Definition:** Neutralization is the chemical reaction that occurs when an **acid and a base** react to form **salt and water only**.

- The acid "cancels out" the base, and the resulting solution is "neutral" (neither acidic nor basic).
- **General Equation:** $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$
- **Examples:**
 1. $\text{HCl (Acid)} + \text{NaOH (Base)} \rightarrow \text{NaCl (Salt)} + \text{H}_2\text{O (Water)}$
 2. An antacid (like milk of magnesia, a base) is used to *neutralize* excess acid in the stomach.
 3. Toothpaste (a base) is used to *neutralize* the acid on your teeth.

Evaluation:

1. What is an acid? List two properties.
2. What is a base? List two properties.
3. What is an indicator?
4. What happens to blue litmus paper in an acid? What happens to red litmus paper in a base?
5. What is a "neutralization" reaction? What are its two products?

Assignment:

1. List three household acids and the specific acid they contain (e.g., Vinegar - Acetic Acid).
2. List three household bases.
3. What is the difference between a base and an alkali?
4. Copy and complete the table for Litmus and Methyl Orange.
5. Give one real-life example of a neutralization reaction.

WEEK 3: THE HUMAN BODY

Topic: The Human Body

Breakdown (Subtopics):

- Digestive, circulatory, and respiratory systems
- Functions of major organs
- Healthy habits and body care
- Simple diagrams and labeling

Performance Objectives: By the end of this lesson, students should be able to:

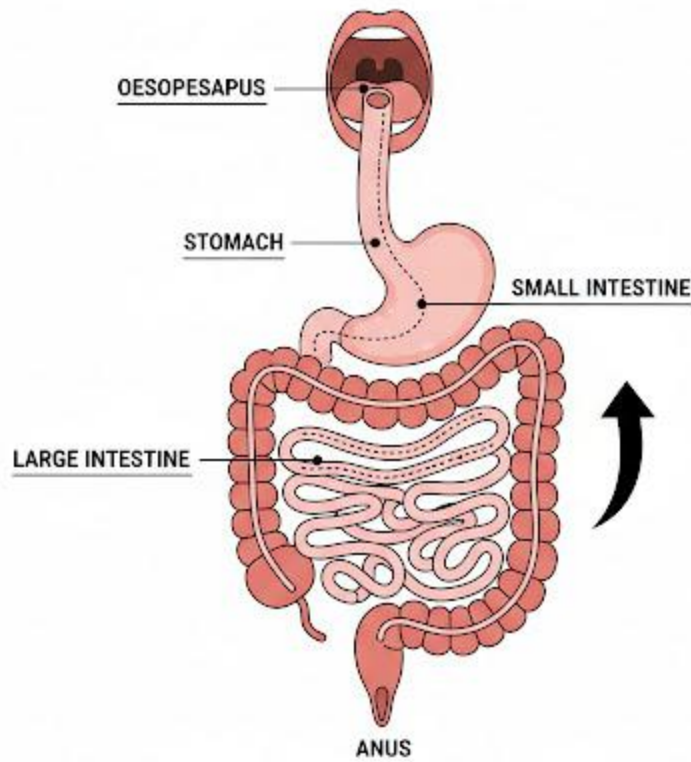
1. Identify the three systems (digestive, circulatory, respiratory).
2. Trace the path of food through the digestive system.
3. State the function of the heart, blood vessels, and blood.
4. Trace the path of air through the respiratory system.
5. List five healthy habits for taking care of the body.

Lesson Content:

1. The Digestive System

- **Function:** To break down large food molecules into small, soluble molecules that can be **absorbed** into the blood.
- **Path of Food (Alimentary Canal):**
 1. **Mouth:** (Digestion begins here). Teeth chew (physical digestion). Saliva contains *amylase* (chemical digestion of starch).
 2. **Oesophagus (Gullet):** A tube that pushes food down to the stomach by muscular waves (peristalsis).
 3. **Stomach:** A muscular bag. It churns food (physical). It releases acid (HCl) to kill germs and *pepsin* (to digest protein).

HUMAN DIGESTIVE SYSTEM



The Pathway of Food Digestion & Nutrient Absorption

4. **Small Intestine:** (Most digestion and absorption occurs here). Enzymes from the pancreas and intestinal wall digest all food (carbs, proteins, fats). The digested food is absorbed through its walls (villi) into the blood.
5. **Large Intestine:** (No digestion). It **absorbs water** from the undigested waste.
6. **Rectum/Anus:** Stores and removes the undigested waste (faeces).

2. The Circulatory System

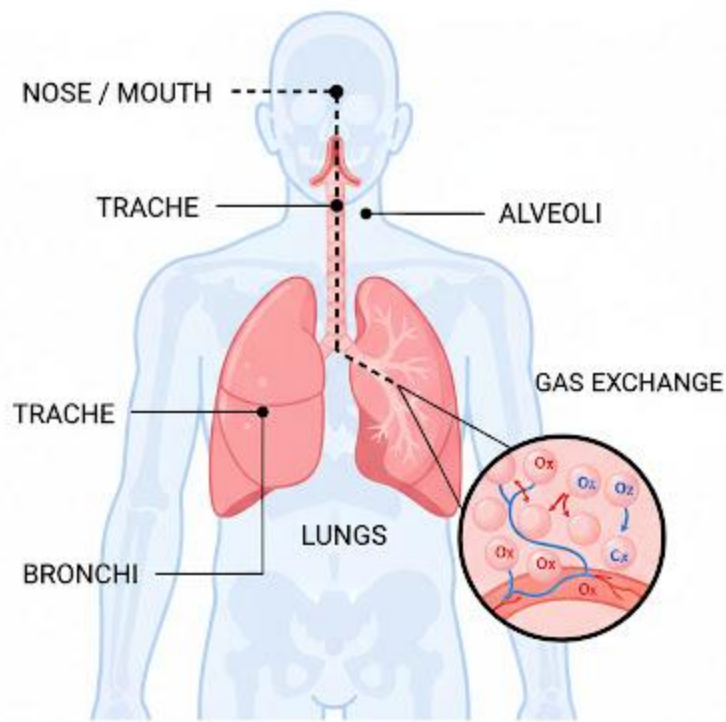
- **Function:** The body's **transport system**. It transports oxygen, digested food, and water *to* all cells, and carries waste products (like carbon dioxide) *away* from them.
- **Major Organs/Components:**

1. **The Heart:** A muscular pump that pumps blood around the body. It has four chambers.
2. **Blood:** The liquid that carries everything. It contains:
 - *Red Blood Cells:* Carry oxygen.
 - *White Blood Cells:* Fight disease.
 - *Platelets:* Help blood to clot.
 - *Plasma:* The liquid part.
3. **Blood Vessels (Tubes):**
 - **Arteries:** Carry blood *Away* from the heart (usually oxygenated, red).
 - **Veins:** Carry blood *back* to the heart (usually deoxygenated, blue).
 - **Capillaries:** Tiny, thin vessels that connect arteries and veins, delivering blood directly to the cells.

3. The Respiratory System

- **Function:** To take in **oxygen** (O₂) and remove **carbon dioxide** (CO₂). This is called *gas exchange*.
- **Path of Air:**
 1. **Nose/Mouth:** Air enters. Hairs and mucus in the nose trap dust and germs.
 2. **Trachea (Windpipe):** The main tube leading to the lungs.
 3. **Bronchi:** The trachea splits into two bronchi (one for each lung).
 4. **Lungs:** The main organs. The bronchi split into smaller tubes (bronchioles), which end in tiny air sacs.
 5. **Alveoli (Air Sacs):** Tiny, balloon-like sacs. This is where **gas exchange** happens. Oxygen passes from the air *into* the blood. Carbon dioxide passes *from* the blood *into* the air to be exhaled.

HUMAN RESPIRATORY SYSTEM



The Pathway of Air & Gas Exchange

4. Healthy Habits and Body Care

1. **Balanced Diet:** Eat from all food groups (carbs, protein, fats, vitamins, minerals, water) to nourish all systems.
2. **Personal Hygiene:** Bathing, brushing teeth, and washing hands to prevent disease.
3. **Regular Exercise:** Strengthens the heart, lungs, and muscles.
4. **Avoid Harmful Substances:** Do not smoke (damages lungs), drink alcohol (damages liver/stomach), or abuse drugs.
5. **Adequate Rest/Sleep:** Allows the body to repair itself.

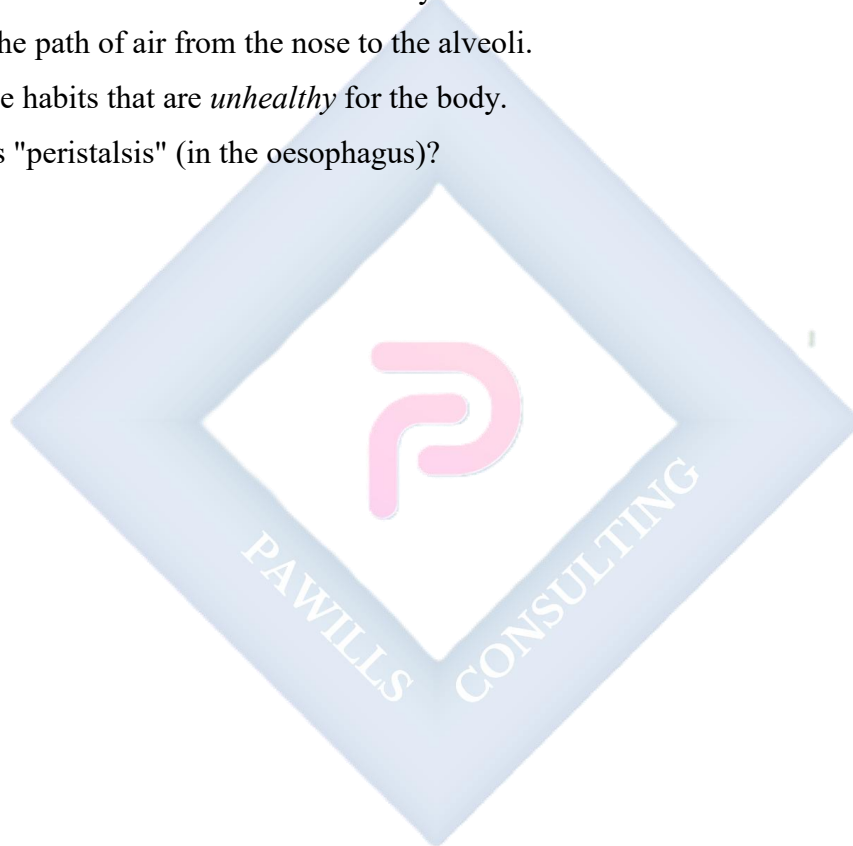
Evaluation:

1. Where does digestion begin in the body?

2. What is the main job of the small intestine?
3. What is the main job of the heart?
4. What is the main job of red blood cells?
5. Where in the lungs does gas exchange (O_2 in, CO_2 out) take place?

Assignment:

1. Draw a simple, labelled diagram of the human digestive system.
2. What is the difference between an Artery and a Vein?
3. Trace the path of air from the nose to the alveoli.
4. List five habits that are *unhealthy* for the body.
5. What is "peristalsis" (in the oesophagus)?

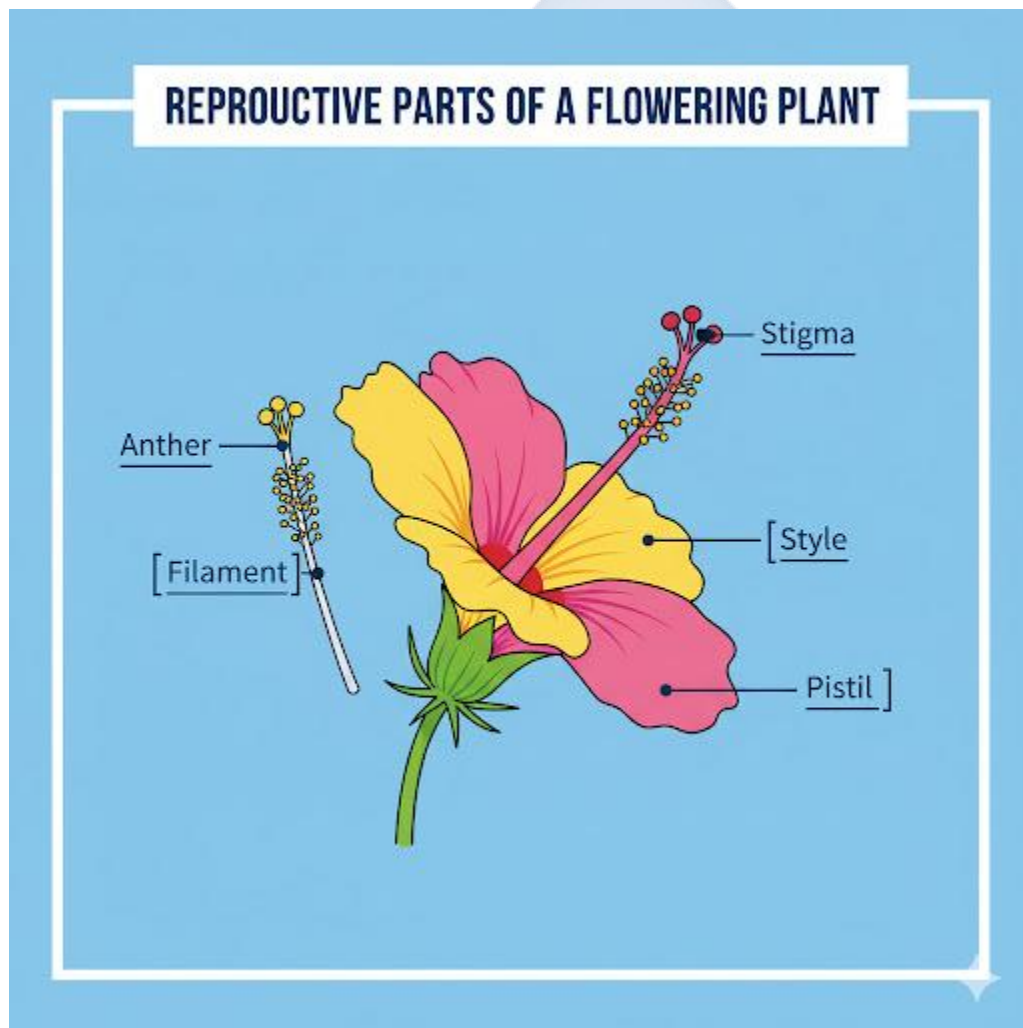


WEEK 4: REPRODUCTION IN PLANTS AND ANIMALS

Topic: Reproduction in Plants and Animals

Breakdown (Subtopics):

- Meaning and types (sexual, asexual)
- Structure and function of reproductive organs
- Importance of reproduction
- Life cycles of selected animals



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Reproduction" and state its importance.
2. Differentiate between Asexual and Sexual reproduction, giving one example of each.
3. Identify the main reproductive parts of a flowering plant (stamen, pistil).
4. Define "Pollination" and "Fertilization."
5. Describe a simple life cycle (e.g., a butterfly).

Lesson Content:

1. Meaning and Importance of Reproduction

- **Definition:** Reproduction is the biological process by which new, individual organisms (called "offspring") are produced from their "parents."
- **Importance:** It is the *only* way for a species to **survive and continue to exist**. Without reproduction, all life on Earth would end.

2. Types of Reproduction

- **A. Asexual Reproduction:**
 - **Definition:** A type of reproduction that involves only **one parent**.
 - **Offspring:** The offspring are genetically **identical** to the parent (they are "clones").
 - **Examples:**
 - **Binary Fission:** (e.g., Amoeba, Bacteria) - One cell splits into two.
 - **Budding:** (e.g., Yeast, Hydra) - A new organism grows off the side of the parent.
 - **Vegetative Propagation:** (in Plants) - A new plant grows from a part of the parent (e.g., planting a yam tuber, cassava stem, or potato).
- **B. Sexual Reproduction:**
 - **Definition:** A type of reproduction that involves **two parents** (a male and a female).
 - It involves the fusion of two special reproductive cells called **gametes**.
 - Male Gamete = **Sperm** (in animals) or **Pollen** (in plants).
 - Female Gamete = **Egg** (or ovum/ovule).

- **Offspring:** The offspring have a *mixture* of traits from both parents (they are *not* identical). This creates genetic variation.
- **Examples:** Humans, birds, fish, and *flowering plants*.

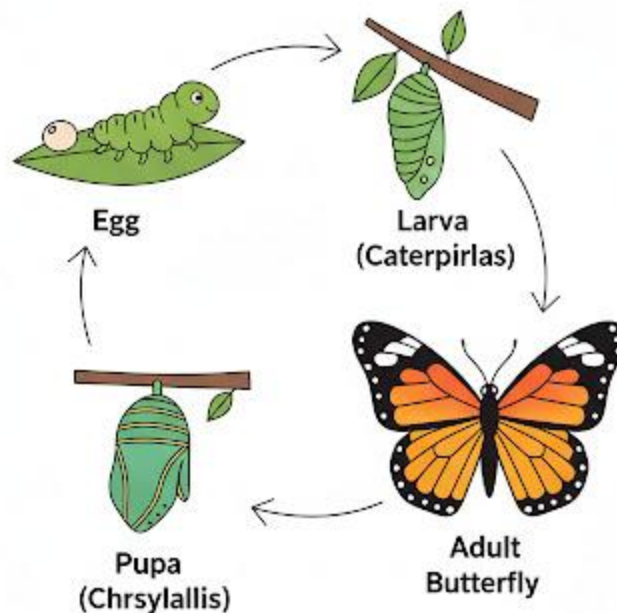
3. Reproduction in Flowering Plants (Sexual)

- **Reproductive Organs:** The flower.
 - **Stamen (Male Part):** Consists of the Anther (produces pollen) and the Filament (holds the anther up).
 - **Pistil/Carpel (Female Part):** Consists of the Stigma (sticky top to catch pollen), Style (tube), and Ovary (contains the ovules/eggs).
- **Process:**
 1. **Pollination:** The transfer of **pollen** from the anther to the stigma. This can be done by wind, insects (like bees), or birds.
 2. **Fertilization:** The pollen grain grows a tube down the style to the ovary. The male gamete from the pollen fuses with the female gamete (ovule).
 3. **Result:** The fertilized *ovule* becomes the **seed**. The *ovary* (which protects the ovules) grows into the **fruit**.

4. Reproduction and Life Cycles in Animals

- **Process:**
 1. The male (e.g., rooster) produces sperm.
 2. The female (e.g., hen) produces eggs.
 3. **Fertilization** (fusion of sperm and egg) occurs, forming a zygote.
- **Life Cycle:** A life cycle shows the stages an organism goes through from birth to adulthood.
 - **Simple Life Cycle (e.g., Human/Bird):** Adult -> Baby (infant) -> Child (chick) -> Adolescent -> Adult
 - **Metamorphosis (e.g., Butterfly):** Egg -> Larva (Caterpillar) -> Pupa (Chrysalis) -> Adult (Butterfly)

LIFE CYCLE OF BUTTERFLY



Four Stages of Complete Metamorphosis

Evaluation:

1. What is the main importance of reproduction?
2. What is the main difference between asexual and sexual reproduction?
3. What is the male gamete (reproductive cell) in a plant?
4. What is "pollination"?
5. List the four stages in the life cycle of a butterfly.

Assignment:

1. Define Asexual and Sexual Reproduction and give two examples of each.

2. Draw a simple flower and label the four main parts (Stamen, Pistil, Petal, Sepal).
3. What is the difference between Pollination and Fertilization in a plant?
4. What does the "ovary" of a flower develop into after fertilization?
5. What does the "ovule" of a flower develop into after fertilization?

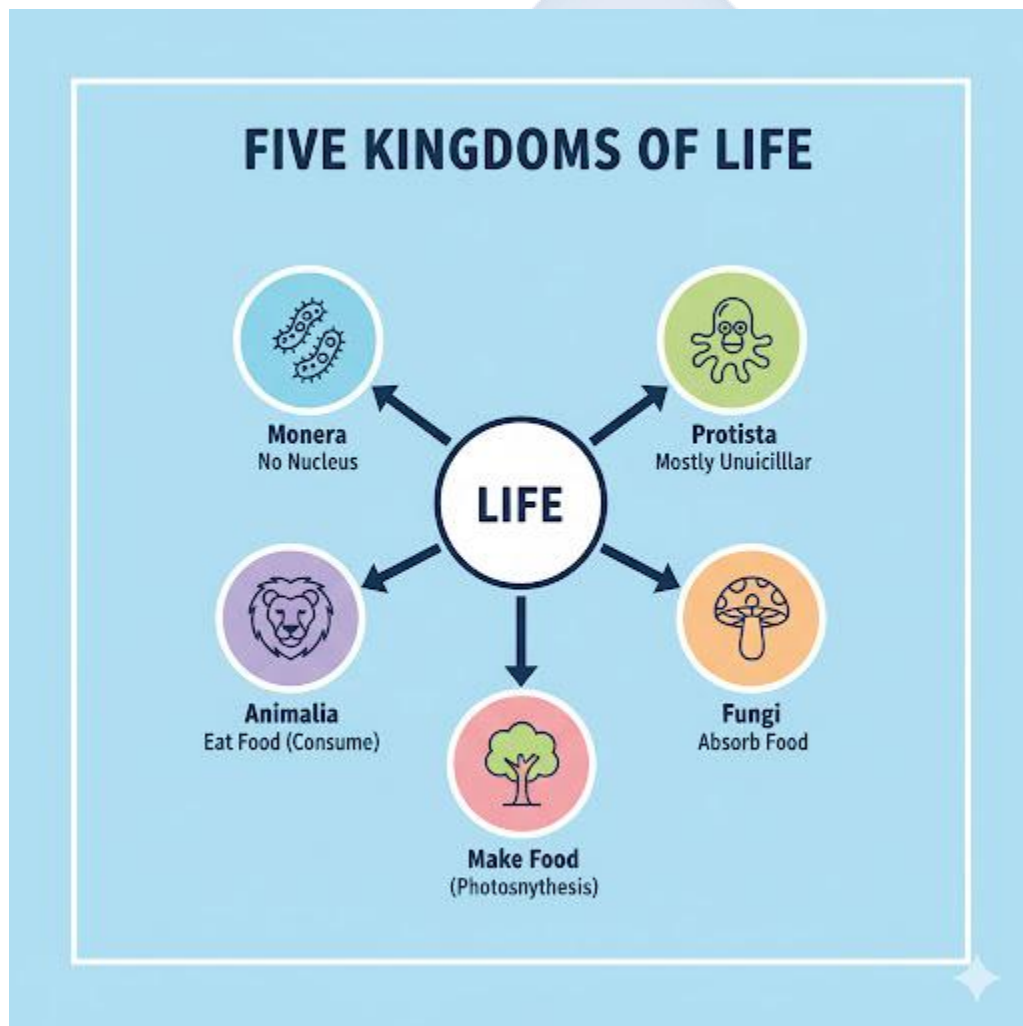


WEEK 6: CLASSIFICATION OF LIVING THINGS

Topic: Classification of Living Things

Breakdown (Subtopics):

- Basis for classification
- Five kingdoms of living organisms
- Major characteristics of each group
- Examples and simple charts



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Classification" (Taxonomy).
2. State two reasons why scientists classify organisms.
3. List the Five Kingdoms of living organisms.
4. State the key (defining) characteristic of each kingdom.
5. Place a common organism (e.g., human, mushroom, mango tree) into its correct kingdom.

Lesson Content:

1. What is Classification?

- **Definition:** Classification (or **Taxonomy**) is the scientific process of **grouping** living organisms based on their **shared characteristics**.
- **Basis (Need) for Classification:**
 1. **Organization:** To organize the millions of living species in a logical and orderly way (like a library organizes books).
 2. **Easy Study:** To make it easier for scientists to study, identify, and name organisms.
 3. **Show Relationships:** To understand how different organisms are related (evolution).
- **History:** The modern system was heavily influenced by **Carolus Linnaeus**, who developed the "Binomial Nomenclature" (two-name system, e.g., *Homo sapiens* for humans).

2. The Five Kingdoms of Living Organisms This is the most common system of classification. All life is divided into five great kingdoms.

- **Kingdom 1: MONERA**
 - **Main Organisms: Bacteria.**
 - **Key Characteristics:**
 1. They are **unicellular** (single-celled).
 2. They are **prokaryotic** (this is the most important feature). A *prokaryote* is a simple cell with **NO true nucleus**.
 - *Example:* Lactobacillus (makes yogurt), E. coli, Salmonella.

- **Kingdom 2: PROTISTA (or PROTOCTISTA)**
 - **Main Organisms:** Amoeba, Paramecium, Algae.
 - **Key Characteristics:**
 1. They are **unicellular** (single-celled).
 2. They are **eukaryotic**. A *eukaryote* is a complex cell that **HAS a true nucleus**.
 - *In short:* They are the "odds and ends" - the eukaryotes that are not plants, animals, or fungi.
- **Kingdom 3: FUNGI**
 - **Main Organisms:** Mushroom, Yeast, Mould (e.g., on bread).
 - **Key Characteristics:**
 1. They are **multicellular** (mostly; yeast is unicellular).
 2. They are **eukaryotic**.
 3. They are **heterotrophs** (consumers) - they get food by *decomposing* dead organisms.
 4. They **do NOT** perform photosynthesis (they are not green).
- **Kingdom 4: PLANTAE (Plants)**
 - **Main Organisms:** Mosses, Ferns, Mango trees, Grass, Flowers.
 - **Key Characteristics:**
 1. They are **multicellular**.
 2. They are **eukaryotic**.
 3. They are **autotrophs** (producers) - they **make their own food** through **photosynthesis** (using chloroplasts).
 4. They are non-motile (cannot move on their own).
- **Kingdom 5: ANIMALIA (Animals)**
 - **Main Organisms:** Insects, Fish, Birds, Lions, Humans.
 - **Key Characteristics:**
 1. They are **multicellular**.
 2. They are **eukaryotic**.
 3. They are **heterotrophs** (consumers) - they must *eat* other organisms for food.

4. They are (mostly) motile (can move on their own).

Evaluation:

1. What is "Classification"?
2. List the Five Kingdoms of life.
3. What is the main, defining difference between Kingdom Monera (Bacteria) and all other kingdoms? (Prokaryotic/No nucleus).
4. Which kingdom do mushrooms and yeast belong to?
5. What is the main difference between Kingdom Plantae and Kingdom Animalia in how they get food?

Assignment:

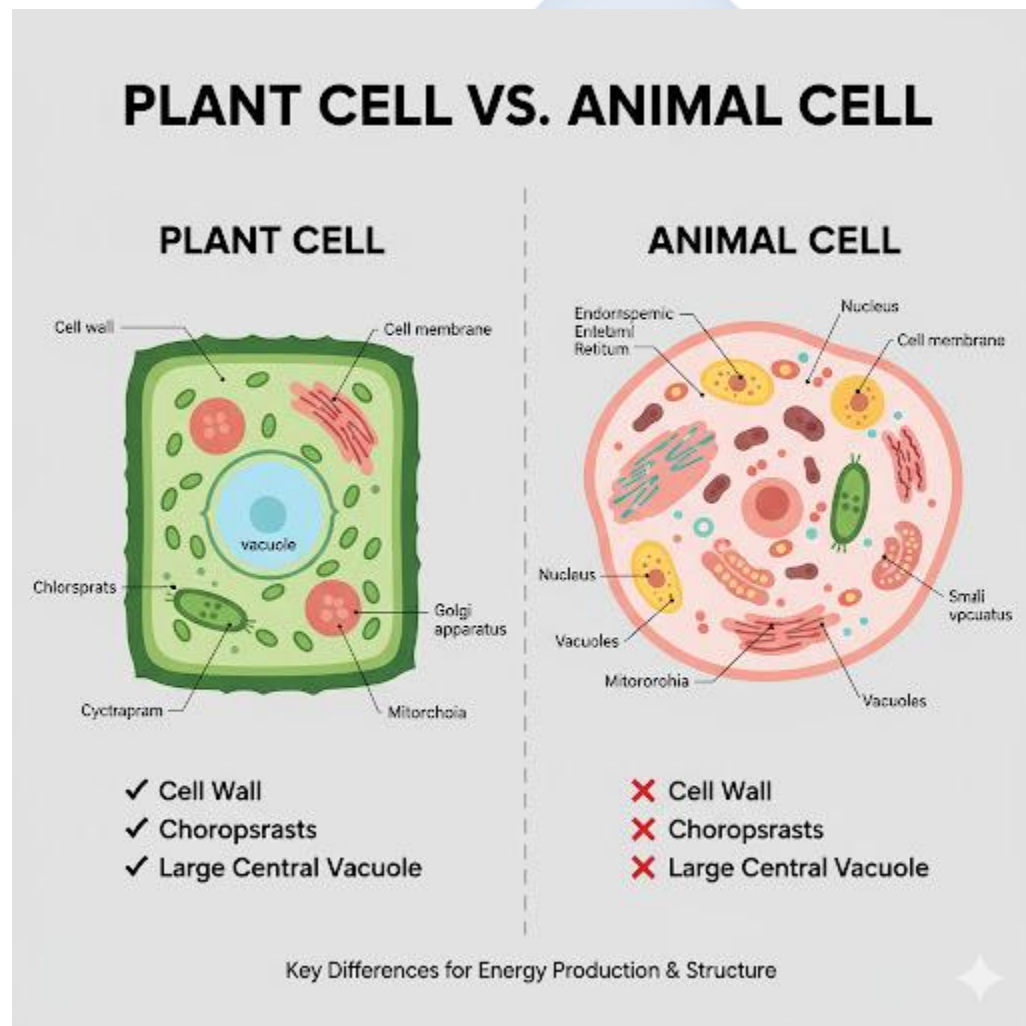
1. List two reasons why classification is important.
2. What is a "prokaryotic" cell? Which kingdom has it?
3. What is a "eukaryotic" cell?
4. In which kingdom would you classify: (a) an Amoeba, (b) a Mango Tree, (c) a Human?
5. Create a simple chart showing the five kingdoms and one key characteristic for each.

WEEK 8: CELLS AS THE BASIC UNITS OF LIFE

Topic: Cells as the Basic Units of Life

Breakdown (Subtopics):

- Meaning and discovery of cells
- Cell structure and functions
- Differences between plant and animal cells
- Use of microscope (demonstration)



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Cell."

2. State the "Cell Theory" (in simple terms).
3. Identify and state the function of four main cell "organelles."
4. List three key differences between a plant cell and an animal cell.
5. State the function of a microscope.

Lesson Content:

1. Meaning and Discovery of the Cell

- **Definition:** The **Cell** is the **basic structural, functional, and biological unit of all living organisms**. It is the smallest unit of life; the "building block" of all life.
- **Discovery:** In 1665, an English scientist named **Robert Hooke** used a simple microscope to look at a thin slice of cork (dead plant bark). He saw small, empty boxes that looked like prison cells, so he called them "**cells**".
- **The Cell Theory:**
 1. All living things are made of **one or more cells**.
 2. The cell is the basic unit of life.
 3. All new cells come from pre-existing cells (cells divide).

2. Cell Structure and Functions (Organelles) Organelles are the tiny "organs" inside a cell that perform specific jobs. *

- **A. Cell (Plasma) Membrane:**
 - **Structure:** The thin, flexible "skin" on the *outside* of an animal cell and just *inside* the cell wall of a plant cell.
 - **Function:** Controls what *enters* (e.g., food, oxygen) and *leaves* (e.g., waste) the cell. It is "semi-permeable."
- **B. Cytoplasm:**
 - **Structure:** The jelly-like substance that fills the entire cell.
 - **Function:** All the other organelles float in it. Many chemical reactions (like metabolism) happen here.
- **C. Nucleus:**
 - **Structure:** A large, (usually) spherical organelle. The "control centre."

- **Function:** It is the "**brain**" of the cell. It controls all cell activities (like growth and reproduction). It contains the **DNA** (genetic material).
- **D. Mitochondrion (plural: Mitochondria):**
 - **Structure:** Bean-shaped organelle.
 - **Function:** The "**powerhouse**" of the cell. This is where **cellular respiration** occurs, releasing energy from food.

3. Plant Cell Structures (The differences) Plant cells have the same four organelles above, PLUS three extra ones:

-
- **E. Cell Wall:**
 - **Structure:** A *tough, rigid* outer layer (outside the cell membrane). Made of **cellulose**.
 - **Function:** Provides strong structural **support** and **protection** for the plant cell. (This is why wood is hard).
- **F. Chloroplasts:**
 - **Structure:** Green, disc-shaped organelles containing **chlorophyll**.
 - **Function:** The site of **photosynthesis** (using sunlight to make food). This is why plants are green.
- **G. Large Central Vacuole:**
 - **Structure:** A very large, central "sac" filled with water, sugar, and salts (cell sap).
 - **Function:** Stores water and helps maintain pressure (turgor) to keep the plant firm. (Animal cells have small, temporary vacuoles, if any).

4. Differences Between Plant and Animal Cells

Feature	Animal Cell	Plant Cell
Shape	Usually irregular / round	Fixed, regular shape (e.g., square)
Cell Wall	Absent	Present (made of cellulose)
Chloroplasts	Absent	Present (in green parts)
Vacuole	Small, temporary, or absent	Large, permanent central vacuole

5. The Microscope

- **Function:** An instrument that uses lenses to **magnify** (make larger) objects that are too small to be seen with the naked eye (like cells, bacteria).

Evaluation:

1. What is a cell?
2. Who first discovered and named the cell?
3. What is the function of the Nucleus?
4. What is the function of the Mitochondrion?
5. List two main differences between a plant cell and an animal cell.

Assignment:

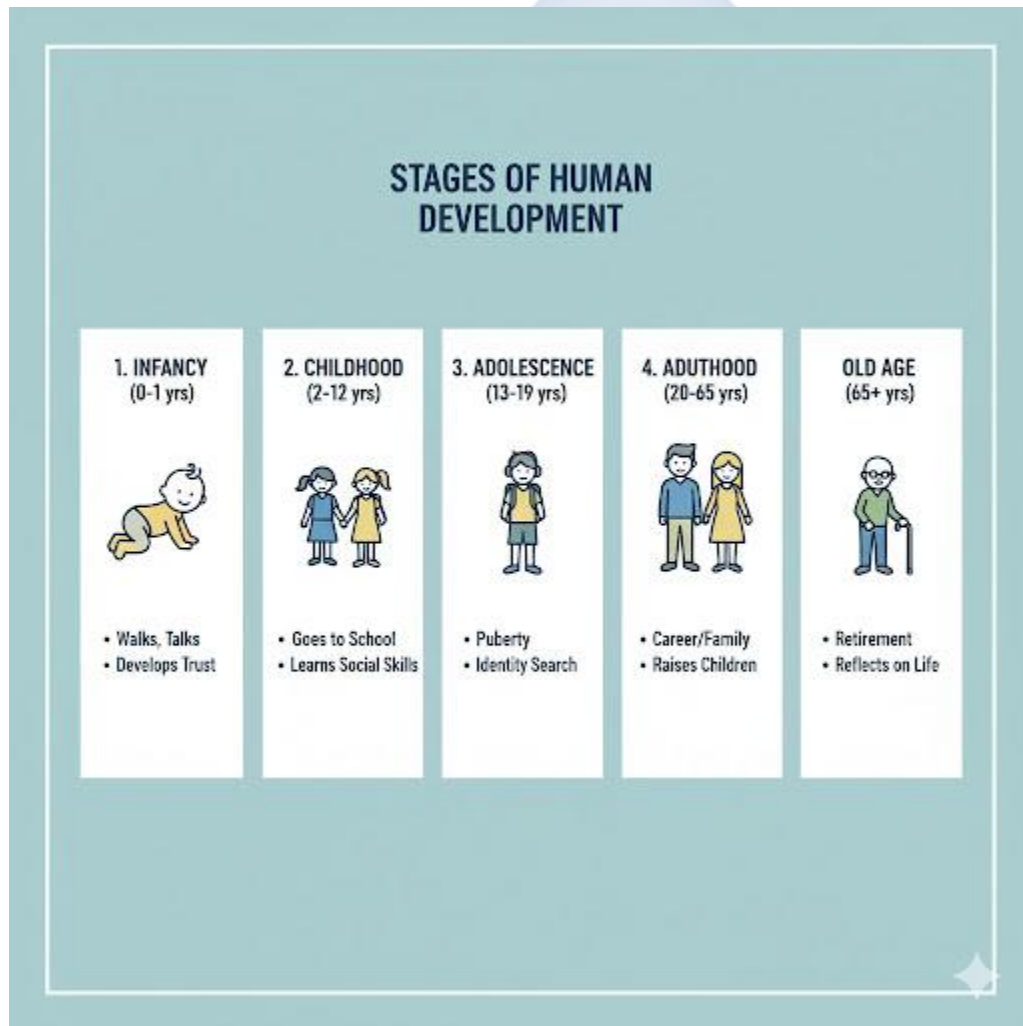
1. State the three parts of the Cell Theory.
2. What is the function of the Cell Membrane?
3. List the three extra organelles that a plant cell has, which an animal cell does not.
4. What is the function of Chloroplasts? What is the name of the green pigment inside?
5. Draw and label a simple plant cell. (Include Cell Wall, Cell Membrane, Nucleus, Cytoplasm, Chloroplasts, Vacuole).

WEEK 9: GROWTH AND DEVELOPMENT

Topic: Growth and Development

Breakdown (Subtopics):

- Definition of growth and development
- Factors affecting growth
- Stages in human development
- Measurement of growth



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Growth."

2. Define "Development."
3. Differentiate between growth and development.
4. List three external and two internal factors that affect growth.
5. List the main stages of human development in order.

Lesson Content:

1. Definition of Growth

- **Definition:** Growth is the **permanent and irreversible increase in the size and mass (weight) of an organism.**
- **How it happens:** It occurs when cells absorb food, get larger, and then *divide* (mitosis) to produce *more* cells.
- **Key Point:** Growth is *quantitative* (it can be measured).

2. Measurement of Growth Growth can be measured using parameters like: ■

1. **Mass (Weight):** Using a weighing scale (measured in kg).
2. **Height (Length):** Using a metre rule or measuring tape (measured in m or cm).
3. **Size:** (e.g., increase in head circumference).

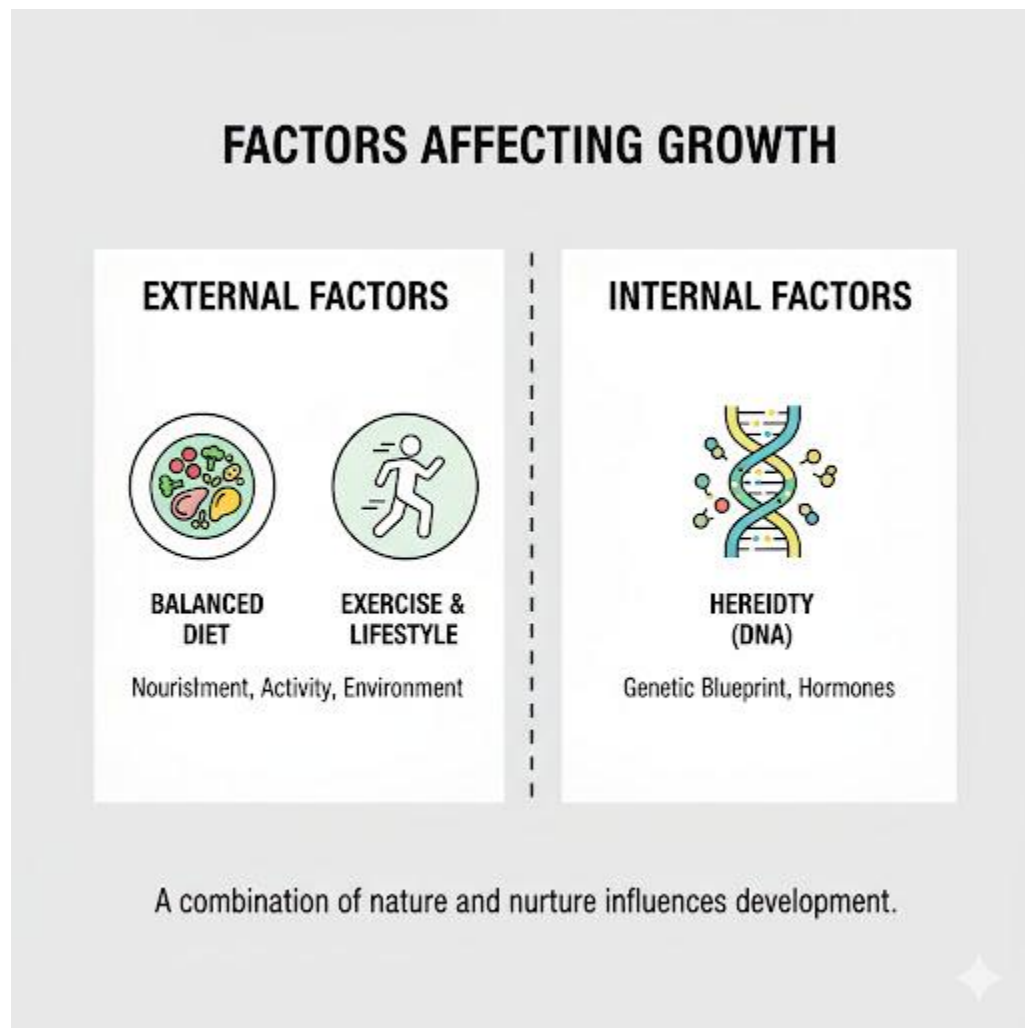
3. Definition of Development

- **Definition:** Development is the **increase in complexity, skill, and functional ability** of an organism.
- **Key Point:** Development is *qualitative* (it is observed as a change in ability).
- *Example:* A baby's brain *grows* (gets bigger), but it also *develops* (learns to talk, walk, and solve problems).

4. Growth vs. Development

- Growth is an increase in **size**.
- Development is an increase in **skill** or **complexity**.

- They are linked, but not the same. A child can *grow* tall but may be *developmentally* delayed (e.g., unable to speak).



5. Factors Affecting Growth For an organism to grow properly, it needs several things:

- **A. External Factors (From the environment):**
 1. **Nutrition (Balanced Diet):** This is the most important. A child needs protein (for building cells), carbohydrates (for energy), and vitamins/minerals (for function). Lack of good food leads to *stunted growth*.
 2. **Exercise:** Stimulates muscle and bone growth.
 3. **Rest/Sleep:** The body releases growth hormones and repairs itself during sleep.
 4. **Clean Environment:** A- disease-free environment.
- **B. Internal Factors (From the parents):**

1. **Heredity (Genes):** The "blueprint" inherited from parents. Genes determine your potential height, body build, etc. (A child with tall parents is likely to be tall).
2. **Hormones:** Chemicals inside the body that control growth (e.g., "growth hormone").

6. Stages in Human Development

1. **Infancy:** (Birth to ~2 years). Period of very rapid growth. Learns to sit, crawl, walk, and say first words.
2. **Childhood:** (2 years to ~10-12 years). Slower, steady growth. Learns to run, talk in sentences, go to school.
3. **Adolescence:** (~12 years to ~19 years). Period of *rapid* growth ("growth spurt"). This is the stage of **puberty**, where the body develops sexually.
4. **Adulthood:** (~20 to 65 years). Growth stops. Body is at its peak, then begins to slowly age.
5. **Old Age:** (65+ years). Body systems slow down, skin wrinkles, bones become brittle.

Evaluation:

1. What is "growth"?
2. What is "development"?
3. Give one example of growth and one example of development in a baby.
4. List two *external* factors that affect growth.
5. List one *internal* factor that affects growth.

Assignment:

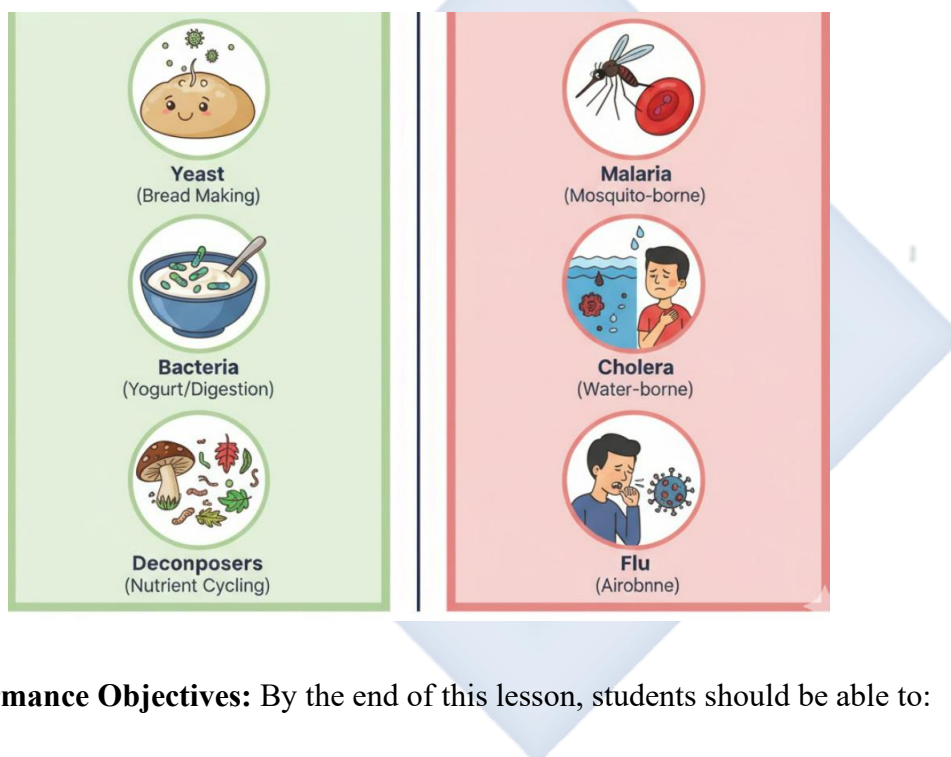
1. What is the main difference between growth and development?
2. List three ways you can measure growth in a human.
3. What is "puberty," and at what stage of development does it occur?
4. Why is a "balanced diet" the most important external factor for growth?
5. List the five main stages of human development in the correct order.

WEEK 10: MICROORGANISMS

Topic: Microorganisms

Breakdown (Subtopics):

- Types: bacteria, fungi, virus, protozoa
- Beneficial and harmful effects
- Control of harmful microorganisms
- Experiment: observing mould growth



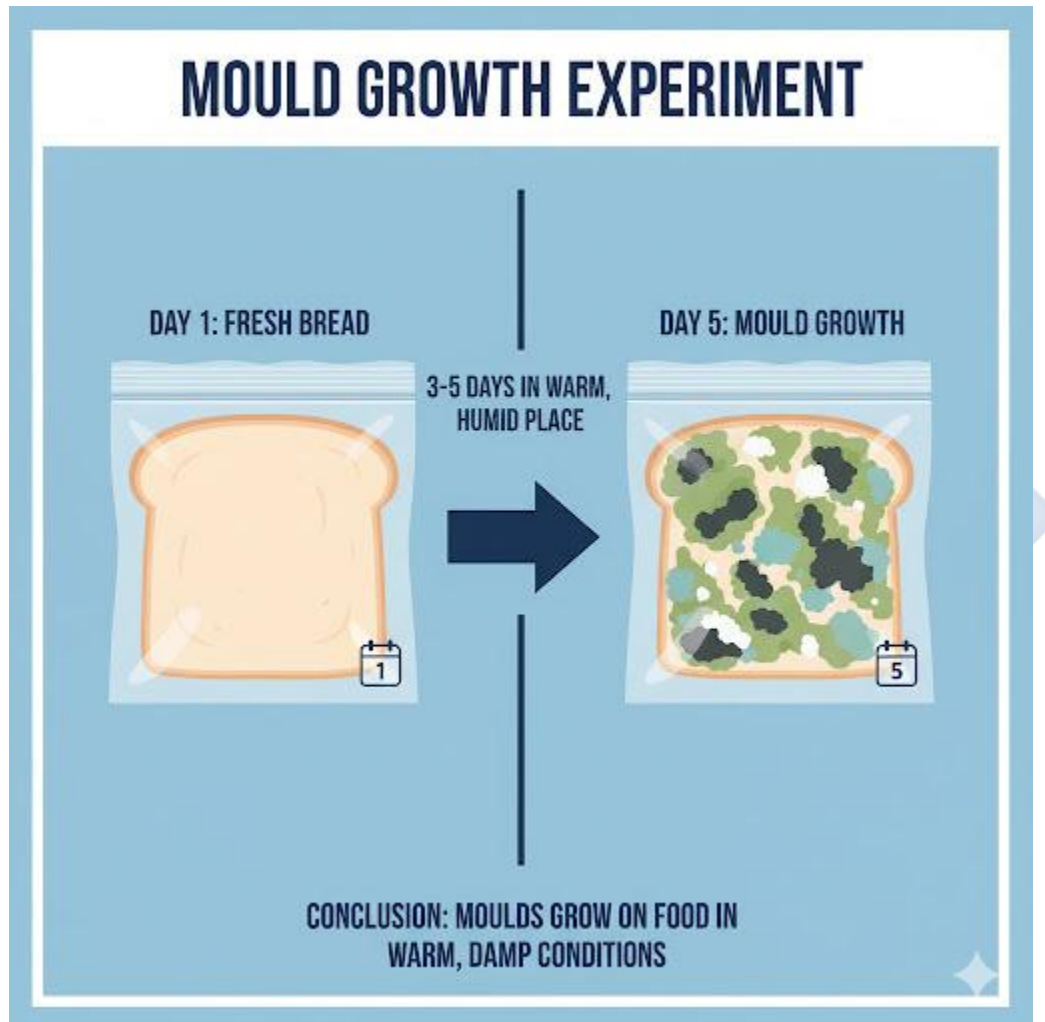
Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Microorganism."
2. List the four main types of microorganisms and give an example of each.
3. List three beneficial (helpful) effects of microorganisms.
4. List three harmful effects of microorganisms.
5. Describe a simple experiment to observe mould.

Lesson Content:

1. What are Microorganisms?

- **Definition:** Microorganisms (or "microbes" or "germs") are living organisms that are **too small to be seen with the naked eye**. They can only be seen with a **microscope**.
- They are found *everywhere*: in the air, in water, in soil, on your skin, and inside your body.



2. Types of Microorganisms

- **A. Bacteria (singular: bacterium):**
 - (Kingdom Monera). Unicellular prokaryotes.
 - *Examples:* Lactobacillus (makes yogurt), Salmonella (causes typhoid), E. coli.
- **B. Fungi (singular: fungus):**

- (Kingdom Fungi). Eukaryotes.
- *Examples:* **Yeast** (unicellular, makes bread rise), **Mould** (multicellular, e.g., on bread), Mushroom.
- **C. Virus:**
 - *Special Case:* Viruses are *not* technically alive. They are just genetic material (DNA/RNA) in a protein coat. They can only reproduce by *invading* a living cell.
 - *Examples:* Influenza virus (causes flu), HIV (causes AIDS), Coronavirus (causes COVID-19).
- **D. Protozoa (singular: protozoan):**
 - (Kingdom Protista). Unicellular eukaryotes.
 - *Examples:* Amoeba, Paramecium, **Plasmodium** (causes malaria, carried by mosquitoes).

3. Beneficial (Helpful) Effects of Microorganisms We could not live without "good" microbes.

1. **Decomposition:** Bacteria and fungi are the main **decomposers** (Week 2). They break down dead plants/animals and rot waste, returning nutrients to the soil.
2. **Food Production:**
 - **Yeast (fungi):** Used in baking (makes bread rise) and brewing (makes alcoholic drinks).
 - **Bacteria:** Used to make yogurt and cheese from milk.
3. **Digestion:** "Good" bacteria live in our large intestine and help us digest food and produce vitamins.
4. **Medicine:** Some fungi and bacteria produce **antibiotics** (e.g., Penicillin from *Penicillium* mould) that kill other harmful bacteria.
5. **Nitrogen Fixation:** Bacteria in the soil "fix" nitrogen from the air into a form that plants can use.

4. Harmful Effects of Microorganisms

1. **Disease:** "Pathogens" are microbes that cause disease in humans, animals, and plants.
 - *Bacteria:* Cause Cholera, Typhoid, Tuberculosis.

- *Fungi*: Cause Ringworm, Athlete's Foot.
 - *Virus*: Cause Flu, COVID-19, AIDS, Measles, Polio.
 - *Protozoa*: Cause Malaria, Dysentery.
2. **Food Spoilage:** Bacteria and fungi cause food (like milk, bread, fruits) to rot, making it unsafe to eat.

5. Control of Harmful Microorganisms

1. **Sanitation/Hygiene:** Washing hands, cleaning surfaces, covering food.
2. **Sterilization (Heat): Boiling** water or **Pasteurization** (heating milk) to kill germs.
3. **Antiseptics/Disinfectants:** Chemicals that kill germs (e.g., Dettol, bleach).
4. **Vaccination:** "Training" your body to fight a virus *before* it attacks you.
5. **Refrigeration:** Using a fridge (low temperature) to *slow down* the growth of microbes on food.

6. Experiment: Observing Mould Growth

- **Aim:** To observe the growth of mould (a fungus) on bread.
- **Method:**
 1. Take a slice of bread.
 2. Sprinkle a few drops of water on it.
 3. Place it in a plastic bag and seal it.
 4. Keep it in a warm, dark place (like a cupboard) for 3-5 days.
- **Observation:** After a few days, black, green, or white "furry" patches will grow on the bread. This is mould.
- **Conclusion:** Microorganisms (like mould spores) are present in the air. They will grow on food if given the right conditions (food, moisture, warmth).

Evaluation:

1. What are microorganisms? What instrument do we need to see them?
2. List the four main types of microorganisms.
3. Which type causes malaria?

4. List two *beneficial* (helpful) effects of microbes.
5. List two *harmful* effects of microbes.

Assignment:

1. Define "pathogen."
2. Give one example of a disease caused by: (a) a bacterium, (b) a virus, (c) a fungus, and (d) a protozoan.
3. What is "pasteurization," and why is it used for milk?
4. What is the role of yeast in baking bread?
5. What is the role of decomposers (bacteria and fungi) in an ecosystem?



WEEK 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Review of all topics in Term 2

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and define the key concepts of the term (chemistry and biology).
2. Explain the differences between elements, compounds, and mixtures.
3. Explain the differences between plant and animal cells.
4. Connect the topics (e.g., cells, growth, reproduction).

Lesson Content:

1. Summary of Term 2 Topics (Chemistry & Biology Term):

- **Week 1 (Chemistry):** Element (pure, one atom, e.g., O), Compound (chemically joined, fixed ratio, new properties, e.g., H₂O), Mixture (physically mixed, any ratio, keeps properties, e.g., salt water).
- **Week 1 (Separation):** Filtration (sand/water), Evaporation (salt/water), Sieving (rice/stones), Magnetism (iron/sand).
- **Week 2 (Acids/Bases):** Acid (sour, litmus red), Base (bitter, soapy, litmus blue), Alkali (base that dissolves in water). Indicator (changes colour). Neutralization (Acid + Base → Salt + Water).
- **Week 3 (Human Body):** Digestive (mouth, stomach, intestines), Circulatory (heart, blood, vessels), Respiratory (nose, trachea, lungs, alveoli).
- **Week 4 (Reproduction):** Asexual (1 parent, identical) vs. Sexual (2 parents, variation). Plant parts (Stamen, Pistil). Pollination (pollen transfer), Fertilization (gamete fusion).
- **Week 7 (Classification):** 5 Kingdoms → Monera (prokaryotic, no nucleus), Protista (eukaryotic, unicellular), Fungi (decomposer), Plantae (producer), Animalia (consumer).

- **Week 8 (Cells):** Basic unit of life. Cell Membrane, Nucleus (brain), Cytoplasm, Mitochondria (power). Plant cells also have: Cell Wall, Chloroplasts, Large Vacuole.
- **Week 9 (Growth):** Increase in *size/mass*. Development is increase in *complexity/skill*. Affected by *nutrition* and *heredity*.
- **Week 10 (Microbes):** (Bacteria, Fungi, Virus, Protozoa). Can be harmful (disease, food spoilage) or beneficial (decomposition, food-making, medicine).

2. Class Activity: Connecting the Dots (The Big Picture)

- How does the Digestive System (Week 3) break down food to provide energy (released by Mitochondria in Cells, Week 8)?
- How do Microorganisms (Week 10, e.g., bacteria/fungi) act as Decomposers (Week 2) in an ecosystem?
- How do Chloroplasts (Week 8) in Plantae (Week 7) make them Producers (Week 2)?
- How is Reproduction (Week 4) and Cell Division (Week 8) essential for Growth (Week 9)?

Evaluation: This week's evaluation is the Second Term Examination.

Assignment:

1. Study all notes from Week 1 to Week 10 for the examination.
2. Draw a table showing 3 differences between a compound and a mixture.
3. Draw a table showing 3 differences between a plant cell and an animal cell.
4. List the 5 Kingdoms and give one example for each.

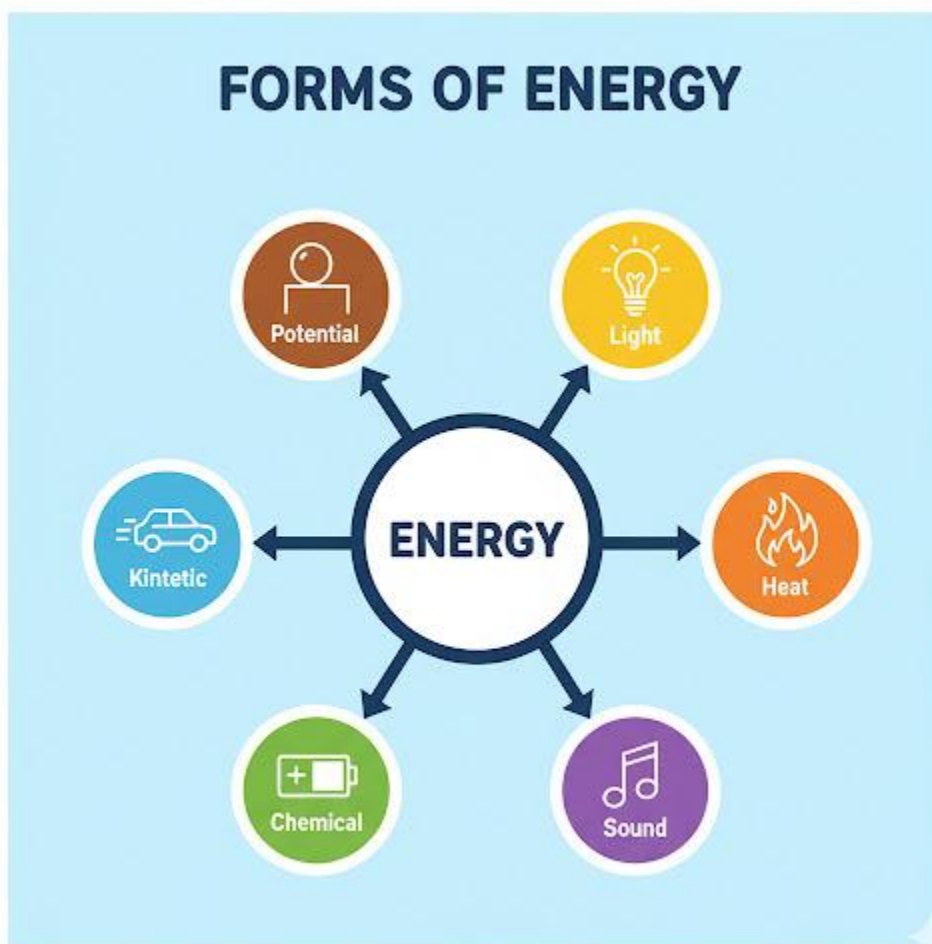
THIRD TERM – JSS 2 INTERMEDIATE SCIENCE LESSON NOTES

WEEK 1: FORMS AND SOURCES OF ENERGY

Topic: Forms and Sources of Energy

Breakdown (Subtopics):

- Light, heat, sound, mechanical, electrical, chemical
- Renewable and non-renewable energy sources
- Energy transformation
- Simple classroom experiments



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Energy" as the ability to do work.
2. List and define at least six different forms of energy.
3. Define "Energy Transformation" and state the Law of Conservation of Energy.
4. Differentiate between renewable and non-renewable energy sources, giving examples of each.
5. Demonstrate simple energy transformations in the classroom.

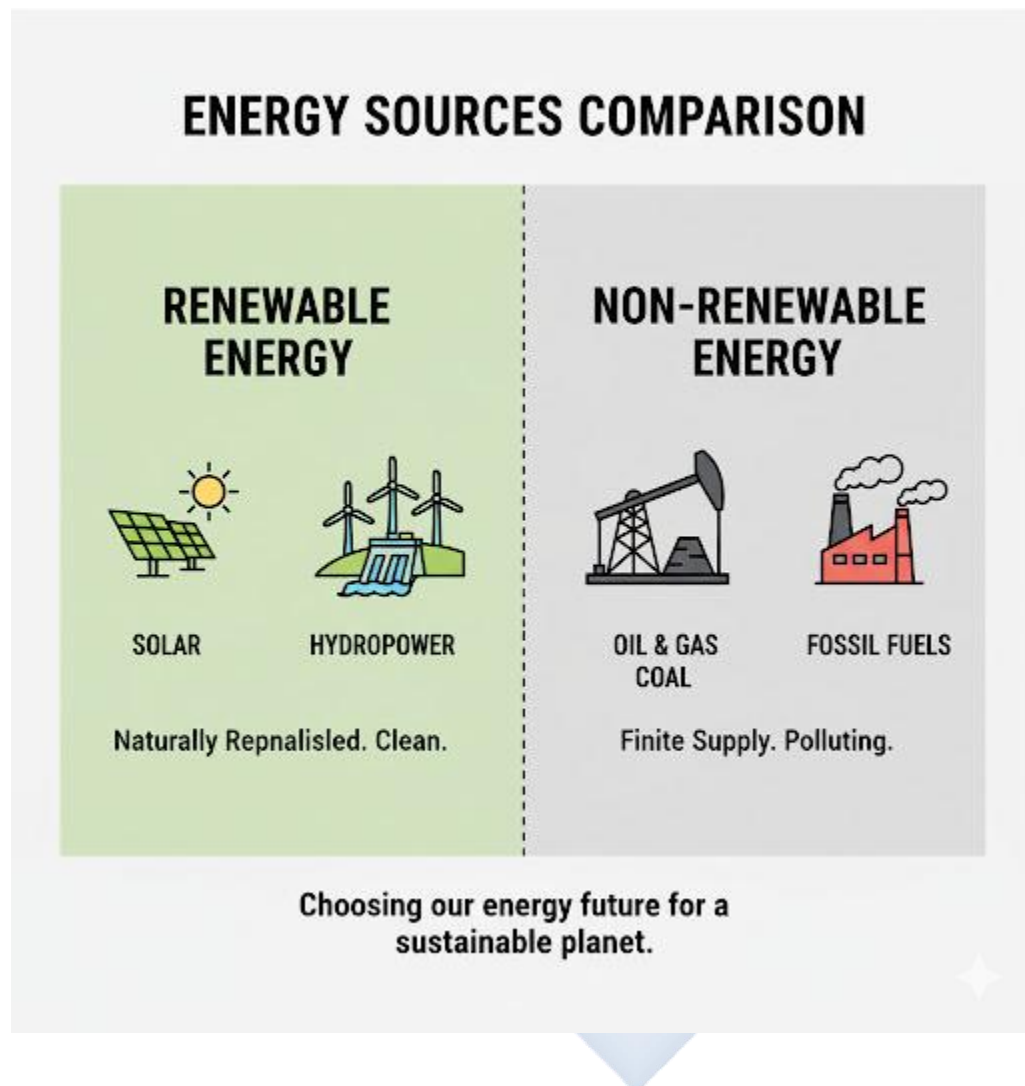
Lesson Content:

1. What is Energy? (Review) Energy is the ability or capacity to do **work** or cause a change. Work is done when a force moves an object over a distance. Energy is measured in **Joules (J)**.

2. Forms of Energy Energy exists in many different forms.

- **A. Light Energy:** The form of energy we can see. It travels as waves.
 - *Examples:* Sunlight, fire, light from a bulb.
- **B. Heat (Thermal) Energy:** The energy of moving or vibrating particles (atoms). It is the energy that flows from a hotter object to a colder one.
 - *Examples:* Fire, a hot stove, friction.
- **C. Sound Energy:** The energy produced by vibrations that travel through a medium (solid, liquid, or gas) and can be heard by our ears.
 - *Examples:* Talking, music, a drum, a clap.
- **D. Mechanical Energy:** The energy of an object due to its *motion* or *position*. It has two types:
 - **Kinetic Energy (KE):** Energy of motion (e.g., a moving car, a thrown ball).
 - **Potential Energy (PE):** Stored energy (e.g., a rock on a hill, a stretched rubber band, a compressed spring).
- **E. Electrical Energy:** The energy from the flow of electric charges (electrons) through a conductor.
 - *Examples:* Electricity from a socket, a battery, lightning.

- **F. Chemical Energy:** Energy that is *stored* in the bonds between atoms and molecules. It is released during a chemical reaction.
 - *Examples:* Food, batteries, petrol (fuel), wood, matches.



3. Energy Transformation and Conservation

- **Energy Transformation:** This is the process of energy *changing* from one form to another.
- **Law of Conservation of Energy:** "Energy can neither be created nor destroyed, but it can be converted (changed) from one form to another."
- **Examples of Transformations:**

- **Electric Bulb:** Electrical Energy → Light Energy + Heat Energy
- **Battery (in a torch):** Chemical Energy → Electrical Energy → Light Energy
- **Human Body:** Chemical Energy (in food) → Mechanical Energy (to move) + Heat Energy (to stay warm)
- **Dropping a ball:** Potential Energy (at the top) → Kinetic Energy (as it falls)

4. Sources of Energy This is *where* we get the energy from.

- **A. Renewable Sources:**

- **Definition:** Energy sources that **can be replaced or regenerated** naturally. They will *not* run out. They are "sustainable" and generally clean.
- **Examples:**
 - **Solar Energy** (from the Sun)
 - **Wind Energy** (from wind)
 - **Hydropower** (from moving water, like Kainji Dam)
 - **Biomass** (from wood, plants, waste)

- **B. Non-Renewable Sources:**

- **Definition:** Energy sources that **cannot be replaced** once used. They are *finite* (limited) and will one day run out. They also cause pollution.
- **Examples:**
 - **Fossil Fuels:** Crude Oil (Petroleum), Natural Gas, Coal
 - **Nuclear Energy** (from minerals like Uranium)

5. Simple Classroom Experiments

1. **Chemical to Heat/Light:** Strike a match. (Chemical energy in the match head converts to heat and light).
2. **Potential to Kinetic:** Hold a ball up (Potential Energy), then drop it (Kinetic Energy).
3. **Mechanical to Heat (Friction):** Rub your hands together fast. They become warm (Mechanical energy converts to Heat energy).

Evaluation:

1. What is energy?
2. List six different forms of energy.
3. What is energy transformation?
4. State the Law of Conservation of Energy.
5. What is the main difference between a renewable and a non-renewable energy source?

Assignment:

1. Define Kinetic Energy and Potential Energy, giving one example of each.
2. List three examples of renewable energy sources and three examples of non-renewable energy sources.
3. Describe the energy transformation that happens in the following:
 - (a) An electric fan
 - (b) A radio
 - (c) A person running a race
4. Why are fossil fuels called "non-renewable"?
5. What form of energy is stored in a battery?

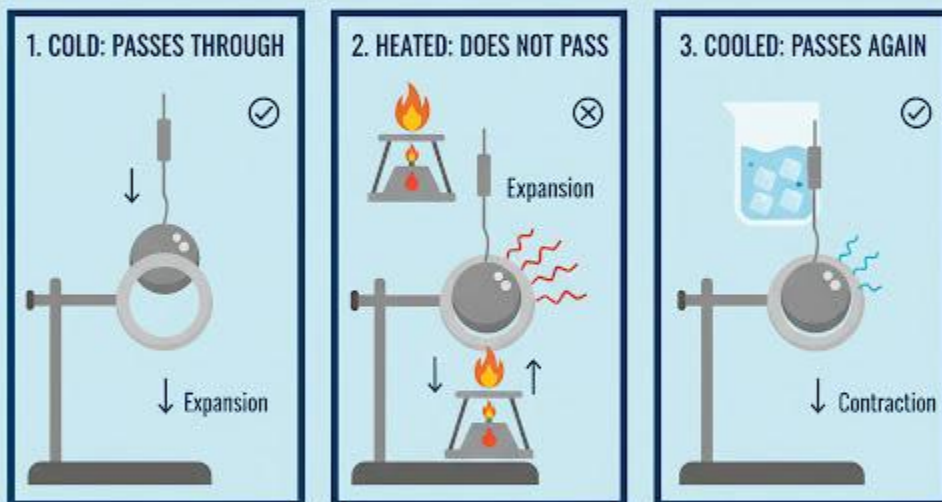
WEEK 2: HEAT AND TEMPERATURE

Topic: Heat and Temperature

Breakdown (Subtopics):

- Meaning of heat and temperature
- Measurement using thermometers
- Effects of heat on matter
- Practical demonstrations

BALL AND RING EXPERIMENT



**CONCLUSION: HEAT CAUSES MATERIALS TO EXPAND,
AND COOLING CAUSES THEM TO CONTRACT.**

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Heat" as a form of energy.

2. Define "Temperature" as the degree of hotness or coldness.
3. Clearly differentiate between heat and temperature.
4. Identify and state the use of a thermometer.
5. List and explain the two main effects of heat on matter (expansion, change of state).

Lesson Content:

1. Meaning of Heat

- **Definition: Heat (or Thermal Energy)** is a form of energy that is transferred from a hotter object to a colder object. It is the *total* kinetic energy of all the particles (atoms/molecules) in a substance.
- **Unit:** Heat is a form of energy, so it is measured in **Joules (J)**.

2. Meaning of Temperature

- **Definition: Temperature** is the **degree of hotness or coldness** of a substance. It is a *measure* of the *average* kinetic energy of the particles in a substance.
- **Unit:** Temperature is measured in **degrees Celsius (°C)** or **Kelvin (K)**.
- **Measurement:** We measure temperature with an instrument called a **Thermometer**.
 - **Laboratory Thermometer:** Used in science for experiments.
 - **Clinical Thermometer:** Used by doctors/nurses to measure body temperature.

3. Differences Between Heat and Temperature

Feature	Heat	Temperature
Definition	A form of energy (total KE).	The <i>degree</i> of hotness (average KE).
Function	It flows from hot to cold.	It is a <i>measurement</i> of hotness.
Instrument	Calorimeter	Thermometer
Unit	Joules (J)	Degrees Celsius (°C) or Kelvin (K)

4. Effects of Heat on Matter When a substance is heated, its particles gain energy and start to move faster and further apart. This causes two main effects:

- **A. Expansion**

- **Definition:** Expansion is the **increase in the size (volume)** of a substance when it is heated.
- **States:**
 - **Solids:** Expand slightly (e.g., metal, like a bridge, expands on a hot day).
 - **Liquids:** Expand more than solids (e.g., the liquid in a thermometer expands and rises).
 - **Gases:** Expand the most (e.g., a balloon in the sun will get bigger).
- **Practical Demonstration (Ball and Ring):**
 - A metal ball just fits through a metal ring when both are cold.
 - Heat the ball: It *expands* and will *not* fit through the ring.
 - Cool the ball: It *contracts* and fits through again.
- **Application:** Gaps (expansion joints) are left in railway tracks and bridges to give them space to expand in the hot sun, otherwise they would buckle and break.

- **B. Change of State**

- **Definition:** When a substance is heated *enough*, it will change its state.
- **The Process:**
 - **Melting:** Solid → Liquid (e.g., Ice → Water)
 - **Evaporation/Boiling:** Liquid → Gas (e.g., Water → Steam)
 - **Condensation:** Gas → Liquid (e.g., Steam → Water)
 - **Freezing/Solidification:** Liquid → Solid (e.g., Water → Ice)
 - **Sublimation:** Solid → Gas *directly* (e.g., Camphor/"Naphthalene balls")

EFFECTS OF HEAT ON STATE CHANGE



Adding Heat Increases Molecular Motion

Evaluation:

1. What is the main difference between heat and temperature?
2. What instrument is used to measure temperature?
3. What is the unit for measuring heat? What is the unit for temperature?
4. What is "expansion"?
5. What is the name of the process when a liquid turns into a gas?

Assignment:

1. Define Heat and Temperature.
2. Draw a simple table showing three differences between heat and temperature.
3. Why are gaps left in railway tracks? (Relate this to "expansion").

4. Define Melting, Boiling, Condensation, and Freezing.
5. What is "sublimation"? Give one example of a substance that sublimates.

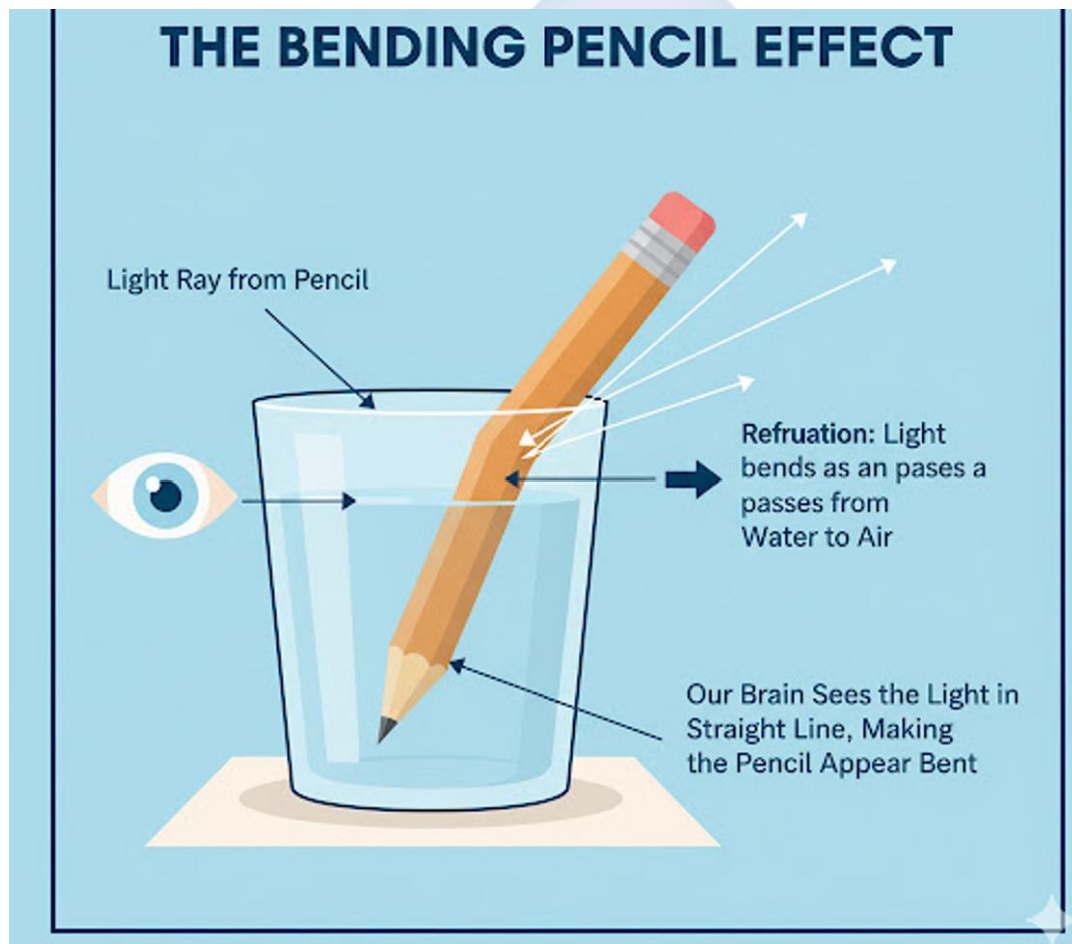


WEEK 3: LIGHT AND ITS PROPERTIES

Topic: Light and Its Properties

Breakdown (Subtopics):

- Reflection, refraction, and absorption
- Transparent, translucent, opaque objects
- Shadows and eclipses
- Simple activities with mirrors and lenses



Performance Objectives: By the end of this lesson, students should be able to:

1. State the main property of light (travels in straight lines).
2. Define Reflection and Refraction, giving one example of each.

3. Differentiate between transparent, translucent, and opaque objects.
4. Explain how a shadow is formed.
5. Define a solar and a lunar eclipse.

Lesson Content:

1. What is Light?

Light is a form of energy (Week 1) that we can see. The main property of light is that it travels in a **straight line** at a very high speed. This is called the **Rectilinear Propagation of Light**.

2. Interaction of Light with Objects

When light hits an object, it can do three things:

1. **Reflection:** The light *bounces off* the object (e.g., a mirror).
2. **Absorption:** The light is *soaked up* by the object (e.g., a black cloth).
3. **Refraction:** The light *passes through* the object but is *bent* (e.g., water, glass).

3. Transparent, Translucent, and Opaque Objects

- **A. Transparent Objects:**

- **Definition:** Objects that allow *all* (or most) light to pass through them easily. You can see clearly through them.
- **Examples:** Clear glass, clean water, air.

- **B. Translucent Objects:**

- **Definition:** Objects that allow *some* light to pass through, but they *scatter* it. You cannot see clearly through them.
- **Examples:** Frosted glass, tissue paper, cloudy water, thin cloth.

- **C. Opaque Objects:**

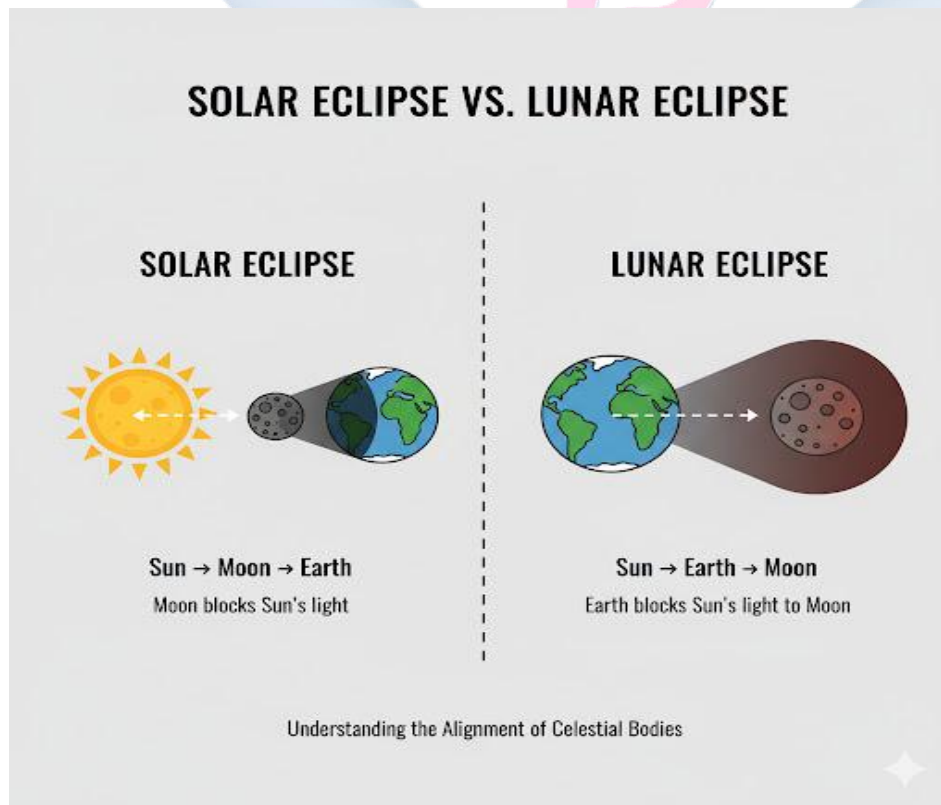
- **Definition:** Objects that do **not** allow *any* light to pass through. They either reflect or absorb all light.
- **Examples:** Wood, stone, metal, a book, your body.

4. Reflection

- **Definition:** Reflection is the *bouncing back* of light when it strikes a surface (usually a shiny one).
- **Mirrors:** A plane (flat) mirror gives a clear, reflected image.
- **Laws of Reflection:**
 1. The angle of incidence is equal to the angle of reflection (Angle i = Angle r).
 2. The incident ray, the reflected ray, and the normal all lie on the same plane.

5. Refraction

- **Definition:** Refraction is the **bending of light** as it passes from one medium (e.g., air) into another medium (e.g., water) at an angle.
- **Cause:** Light bends because it *changes speed* when it enters a new medium. (Light travels slower in water and glass than in air).
- **Practical Example:** A pencil or straw placed in a glass of water appears *bent or broken* at the water's surface.
- **Lenses:** Lenses (like in glasses or magnifying glasses) work by refracting light.



6. Shadows and Eclipses

- **Shadow:** A shadow is formed because light travels in straight lines. When an **opaque object** blocks the path of light, it creates a region of darkness behind it.
 - **Umbra:** The darkest, central part of the shadow.
 - **Penumbra:** The lighter, partial shadow around the umbra.
- **Eclipses:** An eclipse is a large-scale shadow.
 - **Solar Eclipse (Eclipse of the Sun):** The **Moon** passes between the Sun and the Earth. The Moon casts a shadow *on the Earth*. (S-M-E)
 - **Lunar Eclipse (Eclipse of the Moon):** The **Earth** passes between the Sun and the Moon. The Earth casts a shadow *on the Moon*. (S-E-M)

Evaluation:

1. What is the main property of light (how does it travel)?
2. What is the difference between reflection and refraction?
3. What is the difference between a transparent and an opaque object?
4. How is a shadow formed?
5. What is the difference between a solar eclipse and a lunar eclipse?

Assignment:

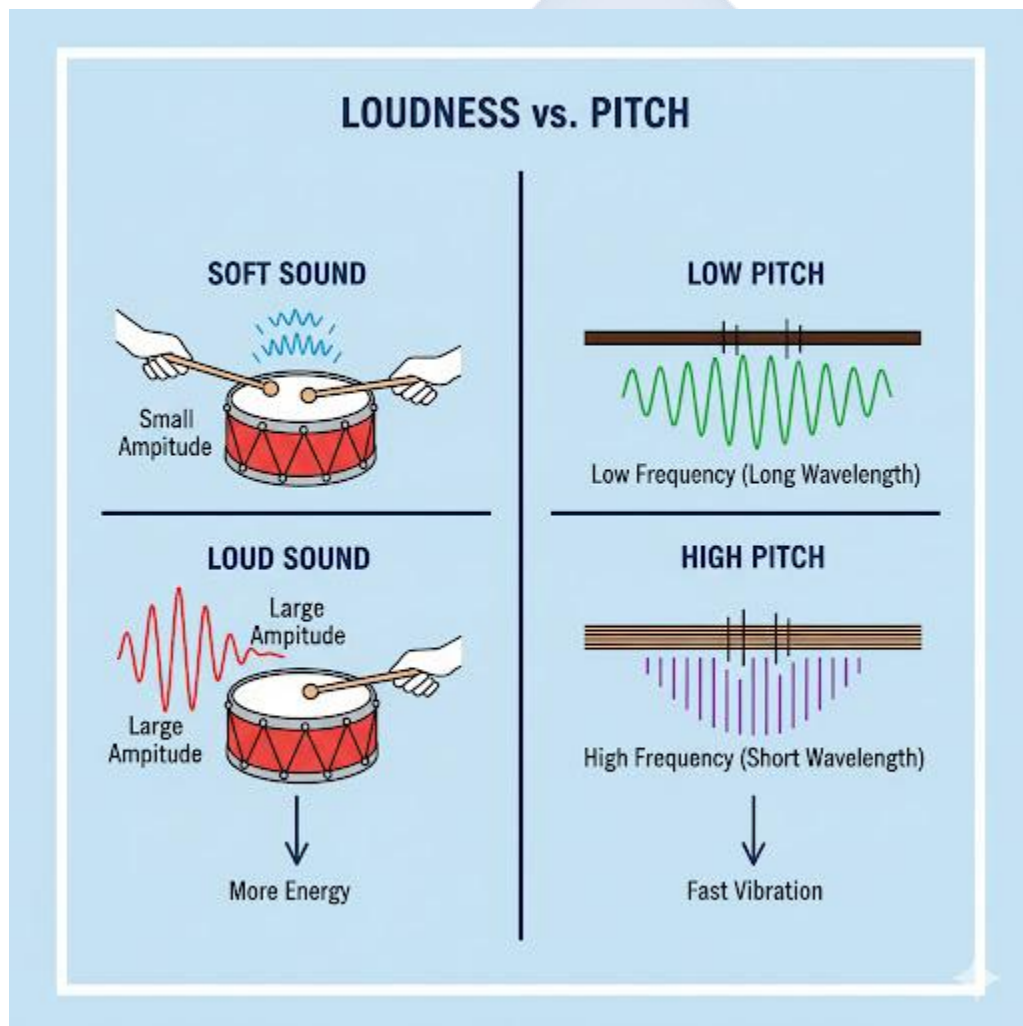
1. Define Reflection, Refraction, and Absorption.
2. List two examples each for Transparent, Translucent, and Opaque objects.
3. Draw a simple diagram showing the "bent pencil" effect of refraction.
4. Draw a simple, labelled diagram showing a solar eclipse (Sun, Moon, Earth).
5. Draw a simple, labelled diagram showing a lunar eclipse (Sun, Earth, Moon).

WEEK 4: SOUND

Topic: Sound

Breakdown (Subtopics):

- Production and transmission of sound
- Loudness, pitch, and vibration
- Noise pollution
- Sound safety measures



Performance Objectives: By the end of this lesson, students should be able to:

1. State that sound is produced by vibrations.

2. Explain that sound needs a medium for transmission.
3. Differentiate between "Loudness" and "Pitch."
4. Define "Noise Pollution" and list two sources.
5. List two sound safety measures to protect hearing.

Lesson Content:

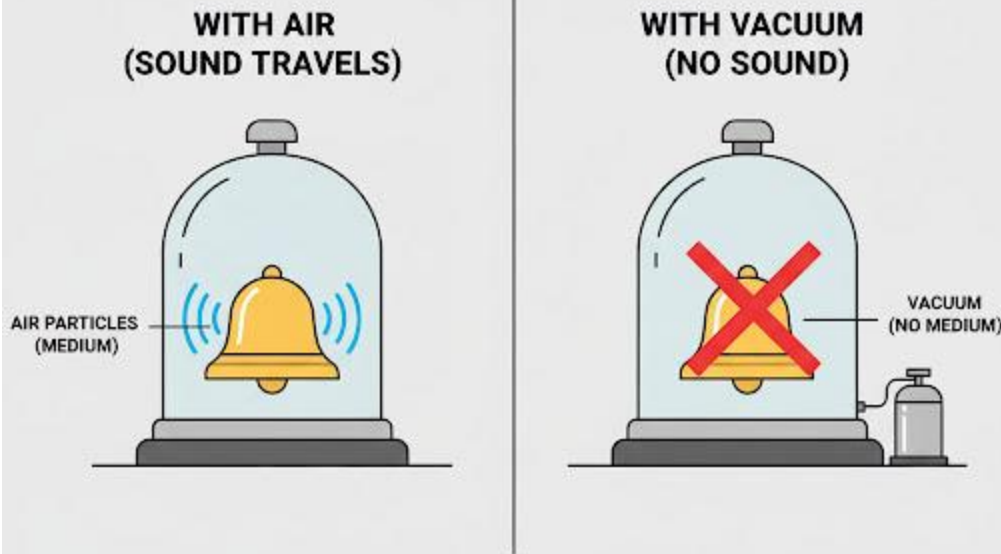
1. Production of Sound

- **Rule:** Sound is produced by **VIBRATIONS**. A vibration is a rapid back-and-forth movement.
- **Examples:**
 - **Guitar:** The string vibrates.
 - **Drum:** The skin (diaphragm) vibrates.
 - **Your Voice:** Your vocal cords (in your throat) vibrate.
 - **Bell:** The metal vibrates.
- *Activity:* Touch your throat and say "Ahhh." You will feel the vibration.

2. Transmission of Sound

- **Rule:** Sound energy travels as waves, but it **MUST have a medium** (a substance) to travel through. The vibrations are passed from one particle to the next.
- **Mediums:**
 - **Solids:** Sound travels *best* (fastest) through solids (e.g., hearing a train on the rail).
 - **Liquids:** Sound travels well through liquids (e.g., dolphins clicking).
 - **Gases:** Sound travels *slowest* through gases (e.g., talking in the air).
- **Vacuum:** Sound **CANNOT** travel in a vacuum (a place with no particles, like outer space).
 - *Bell Jar Experiment:* If you put a ringing bell inside a glass jar and pump out all the air (creating a vacuum), you can *see* the bell ringing but you will *hear no sound*.

SOUND NEEDS A MEDIUM



Sound Requires Particles To Vibrate & Travel.

3. Characteristics of Sound

- **A. Loudness (Volume):**
 - **Definition:** The *intensity* or *strength* of a sound.
 - **Cause:** It depends on the **amplitude** (size) of the vibration. A *harder* drum hit causes a *bigger* vibration and a *louder* sound.
- **B. Pitch (Tone):**
 - **Definition:** The *highness* or *lowness* of a sound.
 - **Cause:** It depends on the **frequency** (speed) of the vibration.
 - **High Frequency (fast vibration) = High Pitch** (e.g., a whistle, a mosquito).
 - **Low Frequency (slow vibration) = Low Pitch** (e.g., a big drum, a cow's moo).

4. Noise Pollution

- **Definition:** (Review from Term 1). Noise is any sound that is **unwanted, unpleasant, or loud enough to cause harm**. Noise pollution is the presence of this sound in the environment.
- **Sources:** Loud generators, car horns, loud music, industrial machinery.
- **Effects:** Hearing loss, deafness, stress, high blood pressure.

5. Sound Safety Measures It is crucial to protect your ears, as hearing damage is often permanent.

1. **Avoid Loud Music:** Do not use earpieces or headphones at a very high volume.
2. **Use Ear Protection:** People who work in noisy places (factories, airports) must wear earplugs or earmuffs.
3. **Distance:** Move away from the source of loud noise.
4. **Avoid Sharp Objects:** Never put sharp objects (like pens or cotton buds) deep into your ear canal.

Evaluation:

1. How is sound produced?
2. What is a "medium"? Why can't sound travel in outer space?
3. What is the difference between "loudness" and "pitch"?
4. What is "noise pollution"?
5. List two ways to protect your hearing.

Assignment:

1. Define "vibration." Give two examples of things that vibrate to make sound.
2. Explain the "Bell Jar Experiment" and what it proves.
3. A mosquito's buzz has a high ____ (pitch or loudness). A lorry's engine is very ____ (pitch or loudness).
4. List three sources of noise pollution in your area.

5. Why is it dangerous to put cotton buds (Q-tips) in your ear?

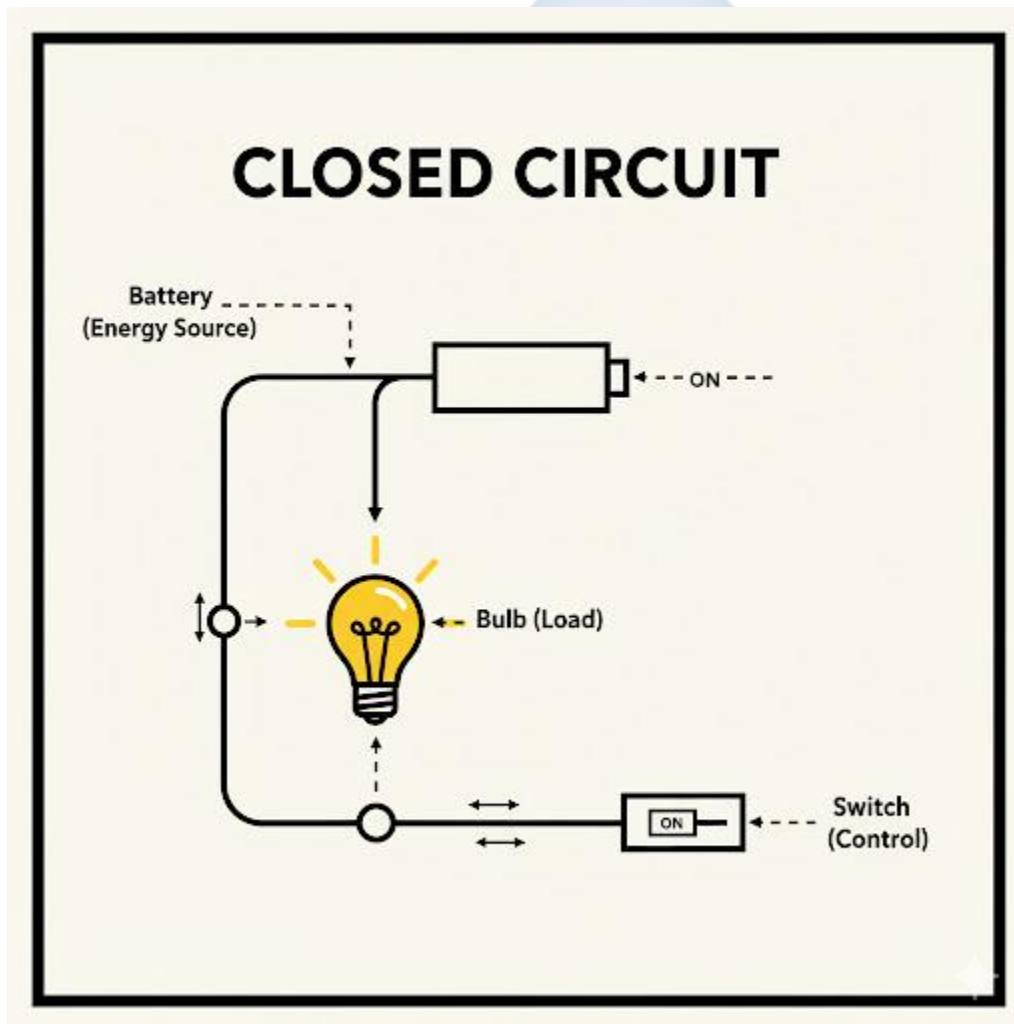


WEEK 6: ELECTRICITY AND MAGNETISM

Topic: Electricity and Magnetism

Breakdown (Subtopics):

- Conductors and insulators
- Simple circuits
- Magnetic and non-magnetic materials
- Safety precautions



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Electricity."

2. Differentiate between a conductor and an insulator, giving two examples of each.
3. Draw and label a simple electric circuit with its four components.
4. Differentiate between a magnetic and a non-magnetic material.
5. List three crucial electrical safety precautions.

Lesson Content:

1. Electricity

- **Definition:** Electricity is a form of energy (Week 1) that results from the flow of electric charges, called **electrons**.
- **Static Electricity:** Charges that *build up* (e.g., rubbing a balloon on your hair).
- **Current Electricity:** Charges that *flow* through a path (e.g., in a wire). This is what we use to power appliances.

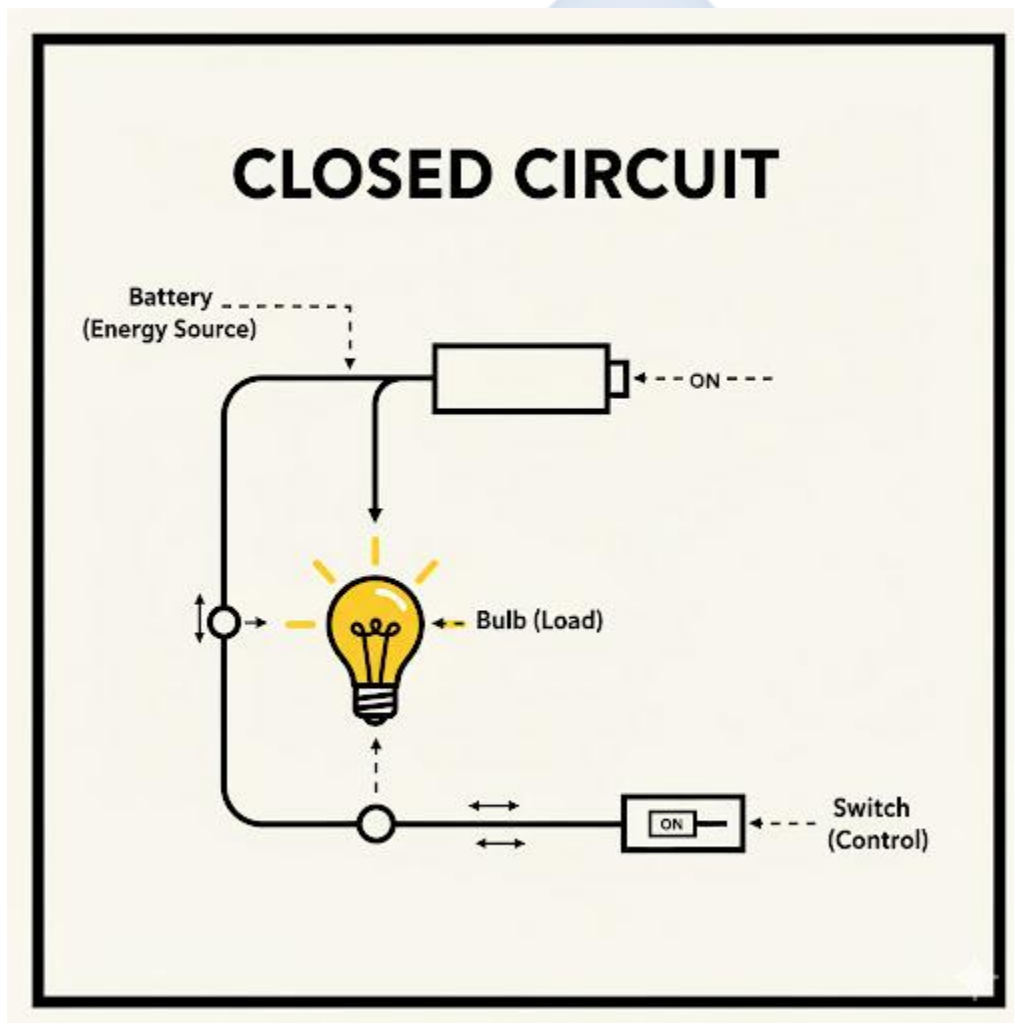
2. Conductors and Insulators

- **A. Conductors:**
 - **Definition:** Materials that allow electricity to pass through them *easily*.
 - **Reason:** Their electrons are loose and free to move.
 - **Examples:** All **Metals** (Copper, Silver, Iron, Gold), Salt Water, Graphite (in a pencil).
- **B. Insulators:**
 - **Definition:** Materials that **block** or *do not* allow electricity to pass through them.
 - **Reason:** Their electrons are tightly held and cannot move.
 - **Examples:** **Rubber, Plastic**, Wood, Glass, Cloth, Pure Water.
 - **Application:** Wires are made of Copper (conductor) on the inside and coated with Plastic (insulator) on the outside to protect us.

3. Simple Electric Circuits

- **Definition:** A circuit is a complete, closed **path** for electricity to flow.
- **Components:**

1. **Energy Source:** Provides the "push" (e.g., a **Battery** or Cell).
 2. **Conductor:** The path (e.g., a **Wire**).
 3. **Load:** The device that *uses* the electricity (e.g., a **Bulb** or motor).
 4. **Control:** A device to open or close the path (e.g., a **Switch**).
- **Types of Circuits:**
 - **Closed Circuit:** The switch is "ON." The path is complete. The bulb *will light up*.
 - **Open Circuit:** The switch is "OFF." The path is broken. The bulb *will not light up*.



4. Magnetism

- **Definition:** Magnetism is the force of attraction (pulling) or repulsion (pushing) caused by a **magnet**.
- **Magnetic Materials:** Materials that are *strongly attracted* to a magnet.
 - **Examples:** Iron, Steel, Nickel, Cobalt.
- **Non-Magnetic Materials:** Materials that are *not* attracted to a magnet.
 - **Examples:** Wood, Plastic, Rubber, Glass, Aluminium, Copper.
- **Poles of a Magnet:**
 - Every magnet has two poles: a **North pole (N)** and a **South pole (S)**.
 - **Law of Magnetism:**
 - Like poles Repel (N-N repel; S-S repel)
 - Unlike poles Attract (N-S attract)

5. Electrical Safety Precautions (CRITICAL) Electricity is extremely dangerous and can kill.

1. **NEVER** touch switches, sockets, or any electrical appliance with **wet hands**. (Water conducts electricity).
2. **NEVER** stick your fingers or metal objects (like forks) into a socket.
3. **DO NOT** overload sockets by plugging in too many appliances (this can start a fire).
4. Always replace broken or "naked" (exposed) wires.
5. Turn off all appliances before you go to bed.
6. Do not fly kites or climb trees near high-voltage power lines.

Evaluation:

1. What is electricity?
2. What is the difference between a conductor and an insulator?
3. List the four main components of a simple circuit.
4. What is the Law of Magnetism? (Like poles..., Unlike poles...)
5. List three important electrical safety rules.

Assignment:

1. Define Conductor and Insulator. Give two examples of each.

2. Draw and label a simple, *closed* circuit.
3. What is the difference between a magnetic material and a non-magnetic material? Give one example of each.
4. Why is it dangerous to touch a socket with wet hands?
5. What will happen if you bring the North pole of one magnet near the North pole of another magnet?

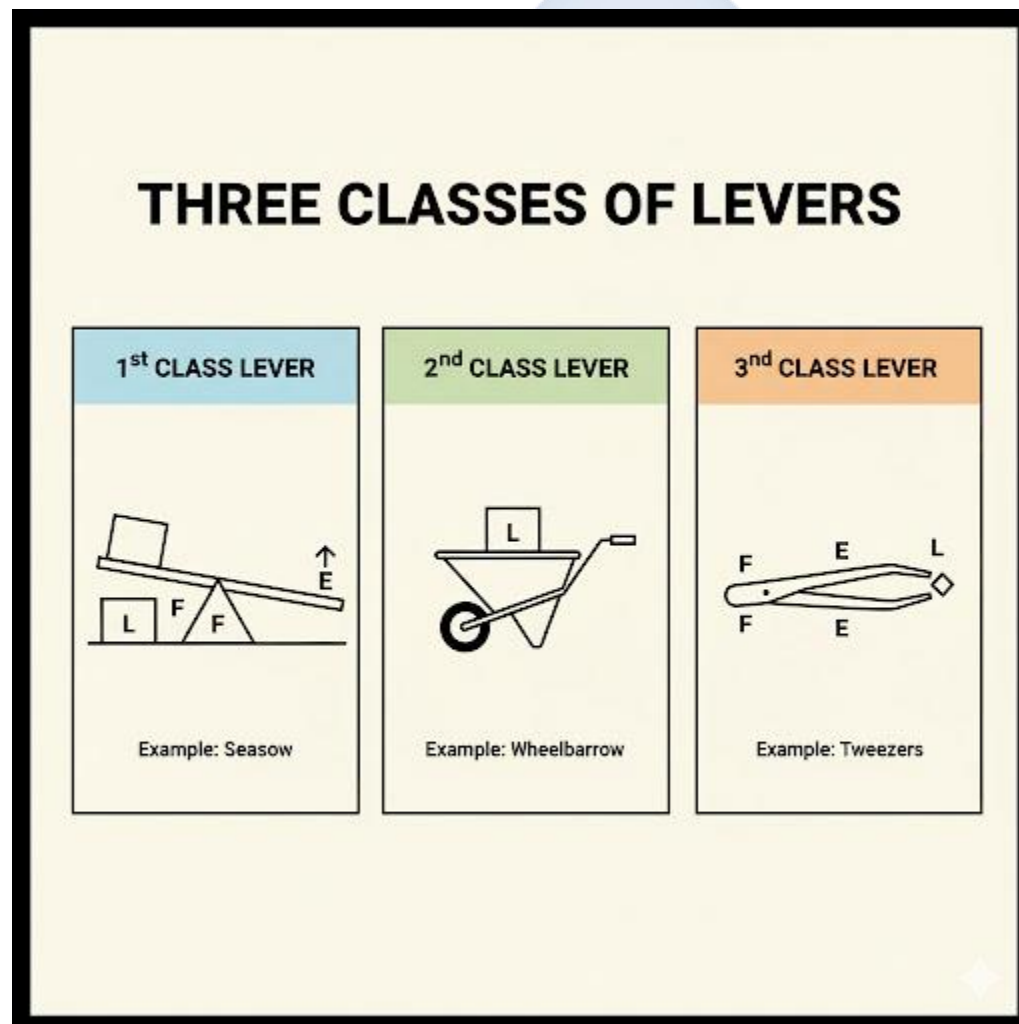


WEEK 8: MACHINES

Topic: Machines

Breakdown (Subtopics):

- Simple machines and their uses
- Levers, pulleys, inclined planes
- Mechanical advantage and efficiency
- Practical demonstrations



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Machine" and "Work" (in the scientific sense).

2. Define "Load" and "Effort."
3. List the six types of simple machines.
4. Identify the three classes of levers with examples.
5. Define "Mechanical Advantage."

Lesson Content:

1. What is a Machine?

- **Work (Scientific):** Work is done when a *force* (a push or pull) moves an object over a *distance*.
- **Machine:** A machine is any device that makes **work easier**.
- **How Machines Help:** They make work easier in three main ways:
 1. **Reducing the Effort:** (e.g., using a lever to lift a heavy stone).
 2. **Changing the Direction of Force:** (e.g., using a pulley to pull a flag *down* to make it go *up*).
 3. **Increasing the Speed:** (e.g., the gears on a bicycle).

2. Load and Effort

- **Load (L):** The *force* that you are trying to overcome (the *weight* of the object).
- **Effort (E):** The *force* that *you* apply to the machine.

3. The Six Simple Machines

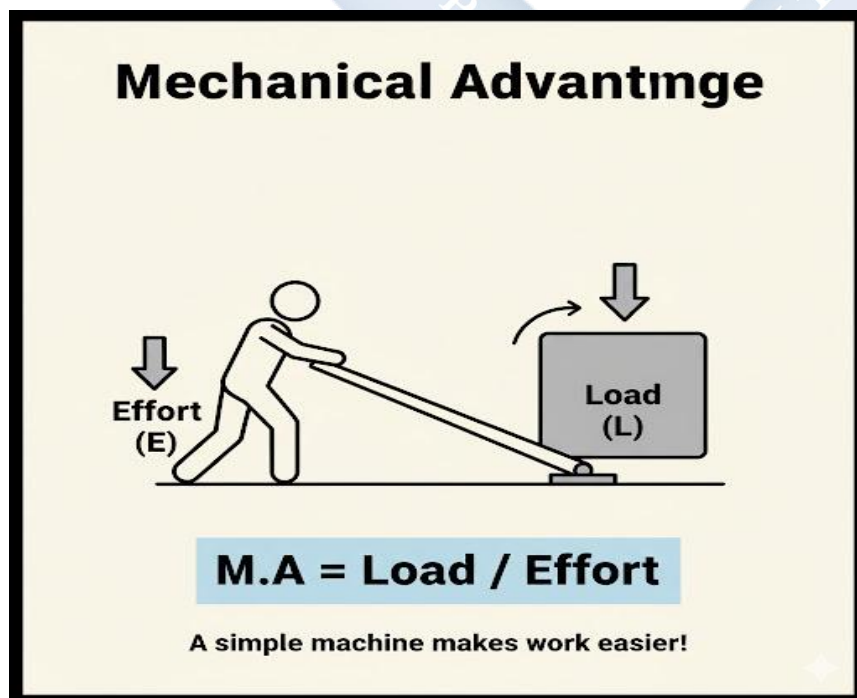
1. **Lever:** A rigid bar that pivots on a fixed point.
2. **Pulley:** A wheel with a groove for a rope.
3. **Inclined Plane:** A flat, sloping surface (a ramp).
4. **Wheel and Axle:** A large wheel fixed to a smaller rod (axle).
5. **Wedge:** (Two inclined planes back-to-back, like a knife or axe).
6. **Screw:** (An inclined plane wrapped around a cylinder).

4. In-Depth: Levers A lever has three parts: **Load (L)**, **Effort (E)**, and **Fulcrum (F)** (the pivot point). The *position* of these three parts determines the class of lever.

- **First Class Lever:**
 - **Arrangement:** Load --- Fulcrum --- Effort (F is in the middle)
 - **Example:** A See-saw, a Crowbar, Scissors (Fulcrum is the screw).
- **Second Class Lever:**
 - **Arrangement:** Fulcrum --- Load --- Effort (L is in the middle)
 - **Example:** A Wheelbarrow (Fulcrum is the wheel, Load is the sand, Effort is your hands), a Bottle Opener.
- **Third Class Lever:**
 - **Arrangement:** Fulcrum --- Effort --- Load (E is in the middle)
 - **Example:** Tongs (Fulcrum is the hinge, Effort is your hand in the middle, Load is the sugar cube at the end), a Fishing Rod, Broom.

5. Inclined Plane and Pulley

- **Inclined Plane:** A ramp. It makes it easier to lift a heavy object by *increasing the distance* (it's easier to roll a barrel up a plank than to lift it straight up).
- **Pulley:** A wheel and rope. A single pulley just *changes the direction* of the force (e.g., at a flagpole).



6. Mechanical Advantage (M.A.)

- **Definition:** This is the *number* that tells you how much the machine multiplies your effort.
- **Formula:** $M.A. = \text{Load} / \text{Effort}$
- **Meaning:**
 - If $M.A. = 5$, it means you only need 10 Newtons of effort to lift a 50 Newton load.
 - If $M.A. > 1$, the machine makes the work easier (reduces effort).

7. Efficiency

- **Definition:** A measure of how much of the work *input* (your effort) is converted to useful work *output* (lifting the load).
- **Formula:** $\text{Efficiency} = (\text{Work Output} / \text{Work Input}) \times 100\%$
- **Fact:** No machine is 100% efficient. Some energy is always lost as **heat** due to **friction**.

Evaluation:

1. What is a machine?
2. What is the "Load" and what is the "Effort"?
3. List the six types of simple machines.
4. What is the arrangement of a First Class Lever? (What is in the middle?)
5. Give one example of a Second Class Lever.

Assignment:

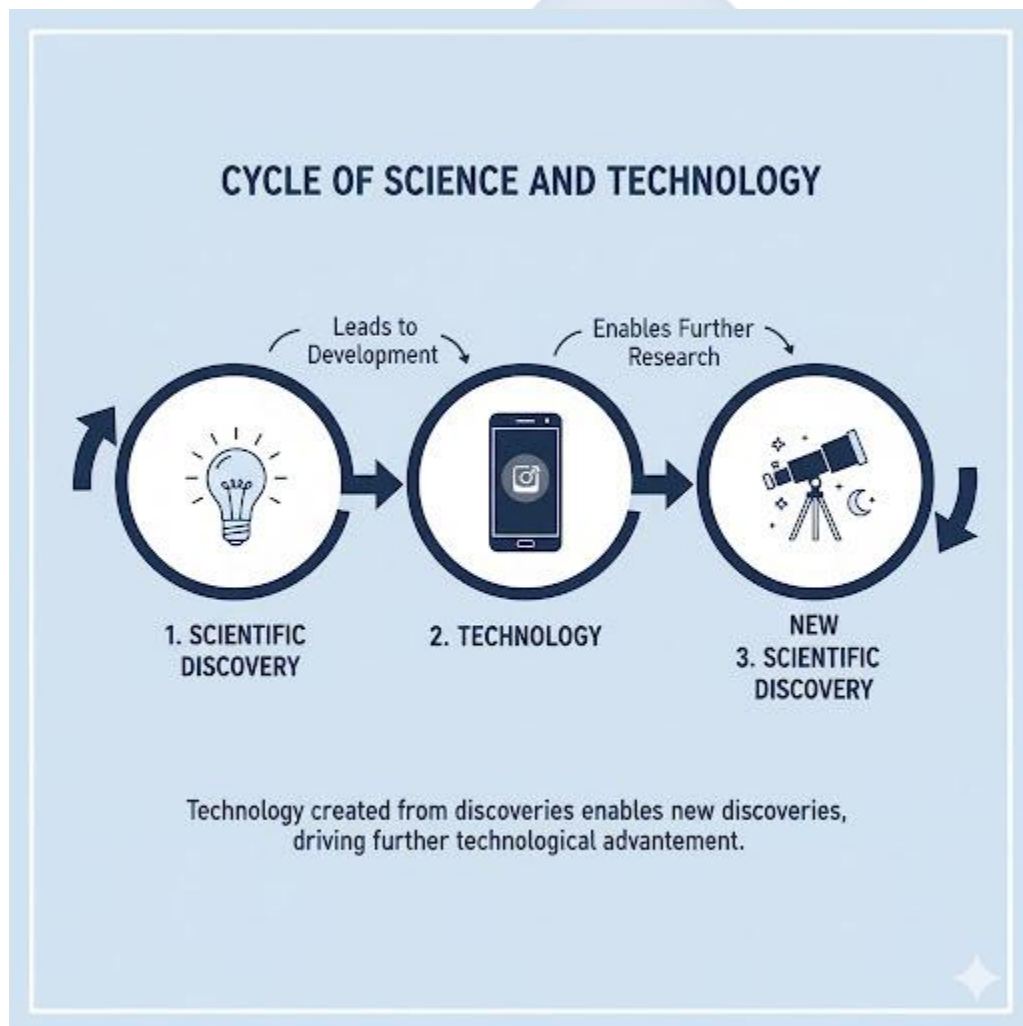
1. Define Mechanical Advantage (M.A.).
2. Define Efficiency. Why is no machine 100% efficient?
3. State the class of lever for each of these: (a) Wheelbarrow, (b) See-saw, (c) Fishing Rod.
4. Draw and label a simple First Class Lever (show the Load, Effort, and Fulcrum).
5. How does an inclined plane (a ramp) make work easier?

WEEK 9: SCIENCE, TECHNOLOGY, AND DEVELOPMENT

Topic: Science, Technology, and Development

Breakdown (Subtopics):

- Relationship between science and technology
- Positive and negative effects of technology
- Careers in science and technology
- National development contributions



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Science" and "Technology."

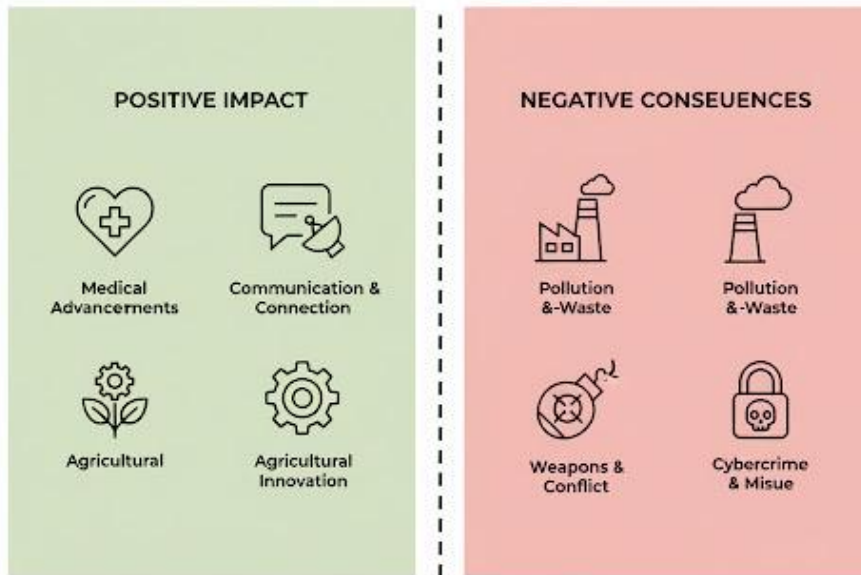
2. Explain the relationship between science and technology.
3. List three positive and three negative effects of technology.
4. List five careers in science and technology.
5. Explain how science and technology contribute to national development.

Lesson Content:

1. Relationship between Science and Technology

- **Science:** (Review) The systematic study of the natural world through observation and experiment. Science is about **knowing why**. (e.g., *Science* is discovering the laws of physics and electricity).
- **Technology:** (Review) The *application* of scientific knowledge to create tools, machines, and processes that solve human problems. Technology is about **knowing how**. (e.g., *Technology* is using the laws of electricity to build a *telephone*).
- **The Relationship (A Cycle):**
 1. Scientific discovery (Science) makes a new invention possible (Technology).
 2. That new invention (Technology, e.g., a telescope) allows scientists to make new discoveries (Science).
 - They depend on each other.

THE DUAL FACE OF TECHNOLOGY



Shaping Our Future Responsibly.

2. Positive and Negative Effects of Technology

- **A. Positive Effects:**

1. **Medicine:** (X-rays, scanners, vaccines, advanced surgery) → People live longer, healthier lives.
2. **Communication:** (Mobile phones, Internet, Satellites) → The world is a "global village."
3. **Agriculture:** (Tractors, irrigation, fertilizers) → More food production.
4. **Transportation:** (Cars, Aeroplanes, Ships) → Faster and easier movement of people and goods.
5. **Education:** (Computers, Internet, projectors) → Easier access to information.

- **B. Negative Effects:**

1. **Pollution:** (Term 1). Cars and factories (technology) cause air, water, and soil pollution.
2. **Job Loss (Automation):** Machines and robots (technology) are replacing human workers in factories.
3. **Weapons:** (Guns, bombs, nuclear weapons) → Technology has made warfare more destructive.
4. **Lifestyle Problems:** (Cybercrime, internet fraud ("Yahoo-Yahoo"), laziness, reduces physical activity).

3. Contributions to National Development

- **National Development** (Review, JSS2 Social Studies, Term 3) is the improvement in the quality of life for all citizens.
- **Role:** Science and Technology are the **main engines** for national development.
 - Good Healthcare (from medical technology)
 - Good Education (from ICT)
 - Economic Growth (from factories, technology, and skilled workers)
 - Good Infrastructure (from engineering technology)
- A nation that ignores science and technology will remain poor and undeveloped.

4. Careers in Science and Technology (S&T)

- **Medicine:** Doctor, Nurse, Pharmacist, Medical Laboratory Scientist.
- **Engineering:** Civil Engineer (builds roads), Electrical Engineer, Chemical Engineer, Software Engineer.
- **ICT:** Computer Programmer, Web Designer, Network Administrator.
- **Pure Sciences:** Biologist, Chemist, Physicist (often work as researchers or teachers).
- **Skilled Trades:** Electrician, Plumber, Mechanic (these are technology roles).

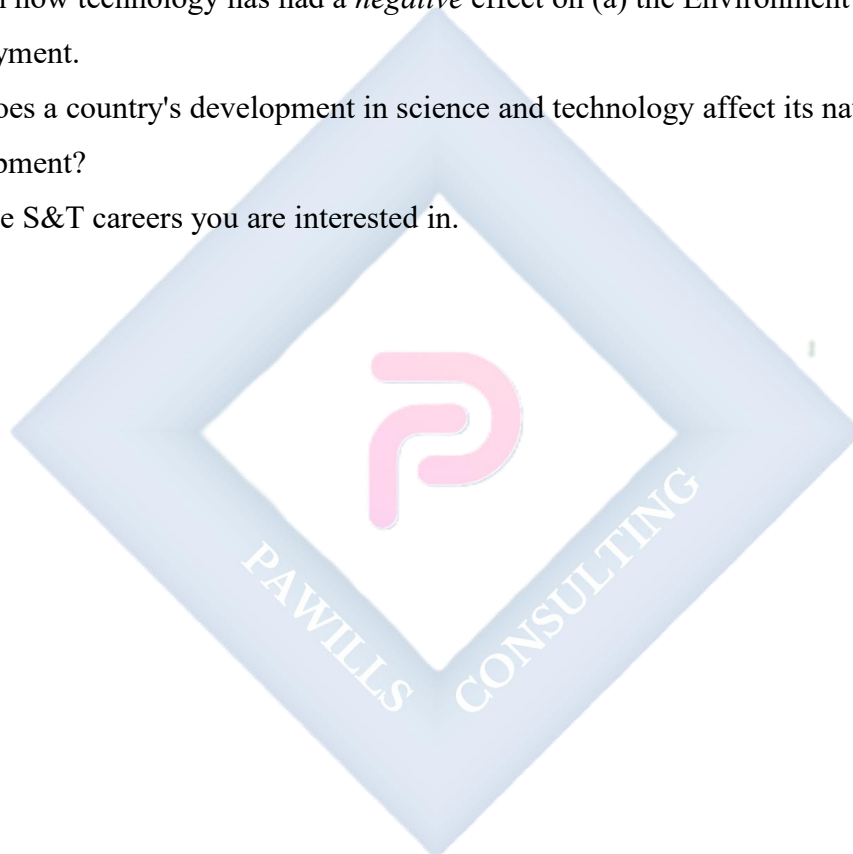
Evaluation:

1. What is the main difference between Science and Technology?
2. Give one example of how science and technology depend on each other.

3. List two positive effects of technology.
4. List two negative effects of technology.
5. List three careers in science and technology.

Assignment:

1. Define Science and Technology in your own words.
2. Explain how technology has had a *positive* effect on (a) Education and (b) Medicine.
3. Explain how technology has had a *negative* effect on (a) the Environment and (b) Employment.
4. How does a country's development in science and technology affect its national development?
5. List five S&T careers you are interested in.



WEEK 10: SKILL ACQUISITION AND ENTREPRENEURSHIP

Topic: Skill Acquisition and Entrepreneurship

Breakdown (Subtopics):

- Meaning and importance of skill acquisition
- Vocational and technological skills
- School-based projects
- Self-reliance through practical skills

PATHWAY TO SELF-RELIANCE



Develop skills, create businesses, gain independence, and empower communities.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Skill Acquisition."

2. Define "Entrepreneurship."
3. Differentiate between academic knowledge and vocational skills.
4. State three reasons why skill acquisition is important for self-reliance.
5. List five examples of vocational/technological skills.

Lesson Content:

1. Meaning of Skill Acquisition

- **Academic Knowledge:** What you learn in class (e.g., the definition of a cell, the history of Nigeria).
- **Skill Acquisition:** The process of learning a *practical ability* to do a specific job or task. It is "learning by doing."
- **Vocational/Technological Skills:** These are practical skills for a specific trade or job.
 - **Examples:**
 - Computer Repairs
 - Web Design / Programming
 - Fashion Design (Tailoring)
 - Catering (Cooking/Baking)
 - Electrical/Electronics Repair
 - Plumbing
 - Carpentry (Furniture-making)
 - Hairdressing / Barbing

2. Entrepreneurship (Review)

- **Definition:** Entrepreneurship is the process of using your **skill**, idea, and resources to **start and run your own business**, taking on financial risks in the hope of making a profit.
- **The Link:** Skill Acquisition (learning to repair computers) + Entrepreneurship (opening your own "Computer Repair Shop") = Self-Reliance.

3. Importance of Skill Acquisition and Entrepreneurship In a country with high unemployment, waiting for the government or a company to give you a job is not enough.

1. **Self-Reliance:** This is the most important benefit. It is the ability to **depend on yourself** for an income. A person with a skill (like a plumber) is *self-reliant* and can feed their family. They don't need to wait for a "job."
2. **Job Creation:** An entrepreneur does not just get a job; they *create* jobs for others. A woman who starts a catering business will hire cooks, cleaners, and delivery drivers.
3. **Solves Unemployment:** It is the *main solution* to the problem of unemployment.
4. **Promotes National Development:** (Week 9). It grows the economy, produces local goods, and solves local problems.



4. School-Based Projects

You can start learning skills right now, based on your science classes.

- **From Chemistry (Week 1 & 2):** Learn to make soap (using a base/alkali), Vaseline, or local indicators.
- **From Biology (Week 3, 4, 8):** Start a small school garden to grow and sell vegetables.
- **From Physics (Week 7):** Learn to build a simple circuit, repair a small radio, or connect a plug.

Evaluation:

1. What is "skill acquisition"?
2. What is "entrepreneurship"?
3. What is the difference between academic knowledge and a vocational skill?
4. What does "self-reliance" mean?
5. List three examples of vocational skills.

Assignment:

1. Define Skill Acquisition and Entrepreneurship.
2. State three reasons why skill acquisition is very important in Nigeria today.
3. How does entrepreneurship help to solve the problem of unemployment?
4. "Skill acquisition leads to self-reliance." Explain this statement.
5. List one vocational skill you are interested in learning and explain why.

WEEK 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Review of all topics in Term 3

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and define the key concepts of the term (physics and technology).
2. Explain the relationship between energy, heat, light, sound, and electricity.
3. Connect the concepts of science, technology, skills, and development.

Lesson Content:

1. Summary of Term 3 Topics (The "Physics & Technology" Term):

- **Week 1 (Energy):** Energy = ability to do work. Forms (Heat, Light, Sound, Electrical, Chemical, Mechanical). Sources (Renewable - Sun/Wind; Non-renewable - Oil/Gas). Transformation (changes form).
- **Week 2 (Heat/Temp):** Heat (total energy, Joules). Temperature (degree of hotness, °C). Effects (Expansion, Change of State - melting, boiling, etc.).
- **Week 3 (Light):** Travels in straight lines. Reflection (bouncing off, mirrors). Refraction (bending, lenses, bent pencil). Opaque (blocks light, forms Shadows) vs. Transparent (allows light through). Eclipses (Solar/Lunar).
- **Week 4 (Sound):** Produced by Vibrations. Needs a medium (cannot travel in vacuum). Pitch (frequency/speed) vs. Loudness (amplitude/size).
- **Week 7 (Electricity/Magnetism):** Electricity (flow of electrons). Conductor (allows flow - metal) vs. Insulator (blocks flow - plastic). Simple Circuit (Battery, Bulb, Switch, Wire). Magnetism (N/S poles, like repels, unlike attracts).
- **Week 8 (Machines):** Make work easier. Load vs. Effort. 6 Simple Machines. Levers (1st, 2nd, 3rd class). M.A. (Load/Effort).

- **Week 9 (S&T):** Science = Knowing. Technology = Doing. Technology has positive (medicine, communication) and negative (pollution, weapons) effects. S&T drives National Development.
- **Week 10 (Skills):** Skill Acquisition = learning a practical ability. Entrepreneurship = starting a business with that skill. Leads to Self-Reliance and solves Unemployment.

2. Class Activity: Connecting the Dots (The Big Picture)

- How is Energy (Week 1) the foundation for all other topics this term?
- How does Technology (Week 9) use Electricity (Week 7), Light (Week 3, e.g., fiber optics), Sound (Week 4, e.g., speakers), and Machines (Week 8)?
- How does Skill Acquisition (Week 10) help a person become an Entrepreneur (JSS2 Social Studies) and contribute to National Development (Week 9)?
- How do the Negative Effects of Technology (Week 9) link back to Pollution (Term 1)?

Evaluation: This week's evaluation is the Third Term Examination.

Assignment:

1. Study all notes from Week 1 to Week 10 for the examination.
2. List the 6 forms of energy and one example of each.
3. Draw a table showing 3 differences between a conductor and an insulator.
4. List the 3 classes of levers and give one example for each class.